

4

Wind-Induced Vibration and Aerodynamic Instability

4.1 Preview

Besides mean wind load and aerostatic instability, as discussed in Chapter 3, there are several mechanisms, in various wind speed ranges, that can excite dynamic response and aerodynamic instability of long-span cable-supported bridges. Wind-induced vibration is an important source of loads on bridge structures, whereas wind-induced aerodynamic instability is very dangerous to bridge structures and may cause the bridge to collapse.

This chapter is going to introduce four types of wind-induced vibration and aerodynamic instability problems that all need to be considered in the design of a long-span cable-supported bridge:

- Vortex shedding excitation usually occurs at low wind speeds and low turbulence conditions, but it can cause considerable vibration of the bridge deck. The interaction of the bridge with vortex flow may result in the so-called “lock-in” phenomenon that leads to excessive bridge vibration. Vortex-induced response of the bridge should be controlled to a certain limit to ensure normal operation and to avoid fatigue damage to the bridge.
- Galloping instability is caused by self-excited forces, and it occurs in vertical modes of the bridge deck. Galloping happens when the bridge deck continuously absorbs energy from the incoming wind flow and the vibration becomes divergent. It happens abruptly and can cause the bridge to collapse, so it should be avoided in the design of the bridge.
- Flutter instabilities of several types occur at very high wind speeds for bridge decks, as a result of self-excited aerodynamic forces. Flutter always involves torsional motions and may also involve vertical bending motions so it, too, should be avoided in the design of bridges. It is flutter that caused the collapse of the original Tacoma Narrows Bridge.
- Buffeting excitation is caused by fluctuating forces induced by turbulence. It occurs over a wide range of wind speeds and normally increases monotonically with increasing wind speed. Excessive buffeting may cause fatigue problem in bridge components and affect functionality of the bridge. The buffeting should be considered in the design stage.

4.2 Vortex-Induced Vibration

4.2.1 Reynolds Number and Vortex Shedding

Vortex-induced vibrations are motions induced on a structure which is interacting with an external flow and are produced by vortex shedding of the flow [1]. The phenomenon that vortices are shed alternately from opposite sides of a circular cylinder is one of the classical open-flow problems in fluid mechanics. At very low Reynolds numbers, the streamlines of the resulting flow around the cylinder are perfectly symmetric, as expected from the potential theory [2]. However, as the Reynolds number increases, the flow becomes asymmetric and the so-called Kármán vortex street occurs, as shown in Figure 4.1.

The flow pattern of incompressible flow around a long circular cylinder perpendicular to the flow depends on the Reynolds number. The Reynolds number is an important parameter in all branches of fluid mechanics and is defined as the ratio of fluid inertia forces to viscous forces:

$$Re = \frac{\rho V D}{\mu} \quad (4.1)$$

where:

Re is the Reynolds number;

ρ is the density of the fluid;

V is the velocity of the fluid relative to the cylinder;

D is the cylinder diameter;

μ is the dynamic viscosity of the fluid.

The flow pattern around a cylinder in different ranges of Reynolds number is shown in Figure 4.2 [3]. For a Reynolds number greater than approximately 30, but less than about 5000, regular shedding of vortices from the two sides of the cylinder occurs, forming a Kármán vortex street or vortex trail downstream from the cylinder, as shown in Figure 4.2a. The flow in this case is unsteady but laminar. As the Reynolds number exceeds approximately 5000, the wake downstream of the cylinder becomes turbulent, whereas the flow around the cylinder upstream of the wake remains laminar. The wake becomes more and more turbulent as the Reynolds number increases.

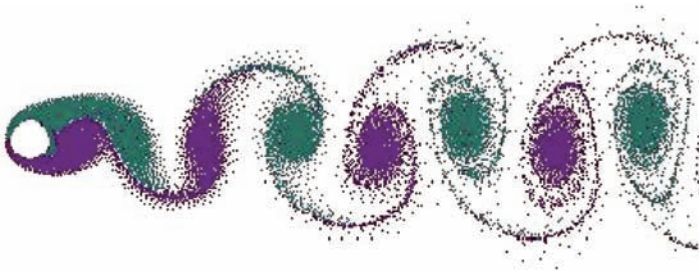


Figure 4.1 Kármán vortex street (Source: Wikipedia (<http://en.wikipedia.org/wiki/File:Vortex-street-animation.gif>), Reproduced with permission from original author: Cesareo de La Rosa Siqueira).

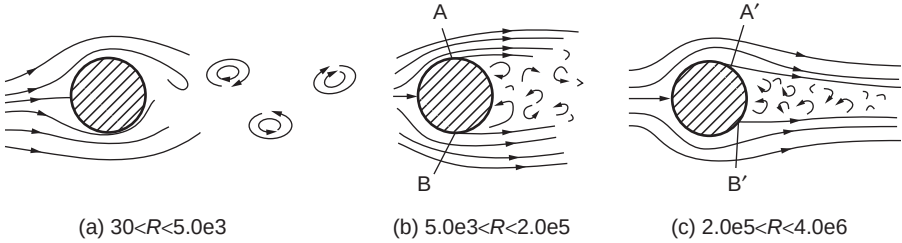


Figure 4.2 Change of flow pattern with Reynolds number.

Before reaching the critical Reynolds number (2×10^5), flow separation occurs on the two sides of the cylinder at points A and B, as shown in Figure 4.2b. The width of the wake is rather wider than the cylinder diameter, and vortex shedding is rather regular. As the Reynolds number exceeds 2×10^5 , the flow separation points suddenly shift downstream from A and B to A' and B', respectively, as shown in Figure 4.2c. This causes a narrower wake and a sudden decrease in drag. The vortex shedding during this stage becomes rather random. Finally, when the Reynolds number exceeds approximately 4×10^6 , the vortex shedding restores some regularity.

From the foregoing discussion, it can be seen that there are three distinctly different ranges of Reynolds number with distinctly different characteristics in vortex shedding, as listed in Table 4.1. Note that in the literature of vortex shedding, the supercritical range is sometimes referred to as critical range, and the hypercritical range is sometimes referred to as transcritical range. The terms used in this chapter are consistent with the use of subsonic, supersonic and hypersonic in studying compressible flow.

4.2.2 Strouhal Number and Lock-In

The Strouhal number, which is named after a Czech scientist, represents the vortex shedding characteristic of a structure and relates the frequency of shedding to the velocity of the flow and a characteristic dimension of the structure (e.g. diameter in the case of a cylinder). Using the Strouhal number, the frequency of vortex shedding can be determined by:

$$f_{st} = S_t \cdot \frac{U}{D} \tag{4.2}$$

Table 4.1 Ranges of Reynolds number and shedding characteristics

Range	Reynolds number	Shedding characteristics
Subcritical	$30 - 2 \times 10^5$	Regular (constant frequency)
Supercritical	$2 \times 10^5 - 4 \times 10^6$	Random (variable frequency)
Hypercritical	$> 4 \times 10^6$	Regular (constant frequency)

where:

f_{st} is the vortex shedding frequency of a structure at rest;

D is the characteristic dimension, usually the across wind width, of the structure;

U is the velocity of the ambient wind flow;

S_f is the Strouhal number.

For a cylinder, the Strouhal number is 0.2 over a wide range of flow velocities.

Assuming that U is slowly increasing from zero, then f_{st} will increase according to Equation 4.2. As each vortex is shed from a structure, a strong crosswind force is induced towards the side of the shed vortex. In this way, the alternate shedding of vortices induces a nearly harmonic cross-wind force. Thus, resonance will first occur when f_{st} becomes equal to the lowest natural frequency of the structure with respect to vibrations in the acrosswind direction or torsion. Further increase of U will cause resonance to occur when f_{st} is equal to the next natural frequency, and so on.

Theoretically, resonance can occur at any natural frequency of the structure [4]. When resonance occurs, the vibration of the structure can be sufficiently large so that the structural vibration can control the frequency of vortex shedding, resulting in a phenomenon known as “lock-in”, as shown in Figure 4.3. The lock-in means that the resonance can sustain through a certain range of wind velocity [5]. Finally, when wind speed is increased to above that causing lock-in, the frequency of shedding will be controlled by the wind again, rather than the structural frequency. The wind speed referring to lock-in is sometimes called the critical wind speed.

4.2.3 Vortex-Induced Vibration

In a long-span cable-supported bridge, the phenomenon of lock-in happens when vortex shedding frequency becomes close to one of the natural frequencies of a bridge component, such as a bridge deck or cable. When it happens, large vibration occurs and becomes an important source of fatigue damage. Therefore, the amplitude of vortex-induced vibrations should be restrained to certain limit in bridge design, and the general mechanism of vortex-induced bridge response should be explored.

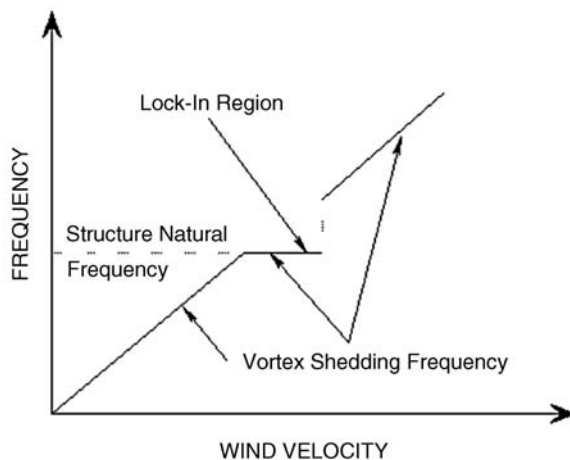


Figure 4.3 Lock-in phenomenon in vortex-induced vibration.

However, because of complex geometrical shapes of structures, the flow patterns around the structures and their relevant responses are also complex. Consequently, vortex-induced vibration has been investigated for simplified structural sections only, such as 2-D rectangular and circular cylinders, and mainly in wind tunnels and in smooth flows.

The governing equation for a structure under vortex-induced load is:

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} = \mathbf{L}_{VS} \quad (4.3)$$

where:

$\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the structural mass, damping, and stiffness matrix, respectively;
 $\mathbf{X}$, $\dot{\mathbf{X}}$, $\ddot{\mathbf{X}}$ are the nodal displacement, velocity, and acceleration response vector; respectively;
 $\mathbf{L}_{VS}$ is the vortex-induced lift force.

At the very beginning, due to the complexity of fluid-structure interaction, simplified analytical models are used to simulate vortex-induced force. Vortex-induced force is assumed to be simple harmonic, as the phenomenon is very similar to a simple harmonic vibration. As a result, the governing equation of vortex induced vibration of a SDOF structure can be written as:

$$m(\ddot{y} + 2\xi\omega_n\dot{y} + \omega_n^2y) = \frac{1}{2}\rho\bar{U}^2DC_L \sin(\omega_s t + \phi) \quad (4.4)$$

where:

m is the mass of the structure;
 y is the vertical displacement;
 ξ is the structural damping ratio;
 ω_n is the structural natural frequency;
 C_L is the lift force coefficient;
 ω_s is the vortex shedding frequency;
 ϕ is the phase angle.

When $\omega_s = \omega_n$, the solution of Equation 4.4 gives the lock-in response of the structure as:

$$y_{\max} = \frac{DC_L\rho\bar{U}^2}{4m\xi\omega_n^2} = \frac{DC_L}{16\pi^2 S_c \cdot S_t^2} \quad (4.5)$$

where $S_c = \frac{m\xi}{\rho D^2}$ is the Scruton number, which is an important non-dimensional structural mass-damping parameter for analysis of wind-induced structural response. Generally speaking, vortex-induced vibrations tend to be mitigated by increasing the Scruton number.

The simple harmonic model of vortex-induced force does not take into account motion-induced forces. One method to consider motion-induced force in modeling vortex-induced force is to assume that the structure is a Van der Pol oscillator. Therefore, the lift force coefficient C_L is time-dependent and should satisfy the following equation:

$$\ddot{C}_L + a_1\dot{C}_L + a_2\dot{C}_L^3 + a_3C_L = a_4\dot{y} \quad (4.6)$$

where a_i ($i = 1 - 4$) are the parameters to be determined by experiments.

A more engineering-oriented model was proposed by Simiu and Scanlan in 1986 [6], in which vortex-induced force can be expressed as:

$$L_{VS}(t) = \frac{1}{2} \rho \bar{U}^2 D \left[Y_1(K) \frac{\dot{y}}{\bar{U}} + Y_2(K) \frac{y}{D} + C_L(K) \sin(\omega_n t + \phi) \right] \quad (4.7)$$

where:

$K = \omega B / \bar{U}$ is the reduced frequency;

$Y_1(K)$, $Y_2(K)$ and $C_L(K)$ are all the functions of the reduced frequency K at lock-in, to be determined by experiments.

A simple harmonic part of the vortex-induced force is included in Equation 4.7 so that the maximum vibration amplitude, which is usually the most important for engineering applications, can be easily calculated. Besides, in the Simiu and Scanlan model, motion-induced stiffness and damping are considered linear.

In 1990, Ehsan and Scanlan [7] revised the above model by adding a non-linear aeroelastic damping coefficient. The revised model can be expressed as:

$$L_{VS}(t) = \frac{1}{2} \rho \bar{U}^2 D \left[Y_1(K) \left(1 - \varepsilon \frac{y^2}{D^2} \right) \frac{\dot{y}}{\bar{U}} + Y_2(K) \frac{y}{D} + C_L(K) \sin(\omega_n t + \phi) \right] \quad (4.8)$$

where ε is the non-linear aeroelastic damping parameter.

The aeroelastic parameters, Y_2 and C_L , are usually ignored, since they have negligible effects on the response of lock-in. Y_1 and ε can be extracted from wind tunnel test results of steady-state amplitudes of the model at lock-in based on the following equation:

$$\frac{y_0}{D} = 2 \left[\frac{Y_1 - 8\pi \cdot S_c \cdot S_t}{\varepsilon \cdot Y_1} \right]^{1/2} \quad (4.9)$$

where:

y_0/D is the reduced amplitude;

y_0 is the vibration amplitude of the vertical displacement;

$S_c = m\xi / \rho D^2$ is the Scruton number;

$S_t = f_{st} D / \bar{U}$ is the Strouhal number;

$f_n = f_{st}$ is the vortex-shedding frequency or the natural frequency.

Once these parameters are obtained, the vortex shedding force acting on the bridge deck per unit length can be obtained by Equation 4.8. The vortex shedding analysis of the bridge can then be performed and vortex shedding-induced response of the bridge can be determined by using Equation 4.3, considering the spatial correlation of vortex shedding-induced forces along the bridge deck and employing the mode superposition method [8].

4.3 Galloping Instability

4.3.1 Galloping Mechanism

“Galloping” is the term used to describe large amplitude vibrations occurring in a direction normal to the mean wind at frequencies much lower than those of vortex shedding from the structure [9].

Galloping is a common instability mode for transmission lines that have non-circular cross-sections due to particular reasons. It can happen to some forms of bridges, usually those with light weight, but it is not a typical instability mode for long-span cable-supported bridges.

Galloping usually occurs at very low reduced frequencies, $B\omega/\bar{U}$, where B is the typical chord or deck width dimension, ω is the frequency of vibration and $\bar{U}$ is the mean free stream velocity. Because the reduced frequency is low, the aerodynamic pressure or force on the bridge deck can be assumed to vary with the incoming velocity in the same manner as for steady flow (the quasi-steady assumption). Therefore, mean or average static aerodynamic data (lift and drag coefficients) can be used to assess the susceptibility of a particular bridge deck to the galloping phenomena.

When a steady wind flow passes an oscillating structure, the effective angle of attack changes with the motion of the structure, even if the incoming flow has a fixed angle of attack. The changing effective angle of attack results in the change of aerodynamic forces and leads to self-excited forces. Consider a 2-D steady flow passing a structure section, as shown in Figure 4.4.

Although the incoming flow velocity U is horizontal, the actual wind velocity acting on the structure is U_α , with an effective angle of attack α , because of the motion of the structure in the y -direction. Based on the quasi-steady assumption, the drag and lift forces can be expressed as:

$$D(\alpha) = \frac{1}{2} \rho U_\alpha^2 B C_D(\alpha) \quad (4.10a)$$

$$L(\alpha) = \frac{1}{2} \rho U_\alpha^2 B C_L(\alpha) \quad (4.10b)$$

where:

$D(\alpha)$ and $L(\alpha)$ are the drag and lift forces on the structure section, respectively;

U_α is the wind velocity with effective angle of attack α ;

$B = 2b$ is the bridge deck width;

$C_D(\alpha)$ and $C_L(\alpha)$ are the drag and lift coefficients of the structure section, respectively.

The vertical force in the y -direction can then be calculated:

$$\begin{aligned} F_y(\alpha) &= \frac{1}{2} \rho U_\alpha^2 B (C_D(\alpha) \sin \alpha + C_L(\alpha) \cos \alpha) \\ &= \frac{1}{2} \rho U^2 B (C_D(\alpha) \tan \alpha + C_L(\alpha) \sec \alpha) \end{aligned} \quad (4.11)$$

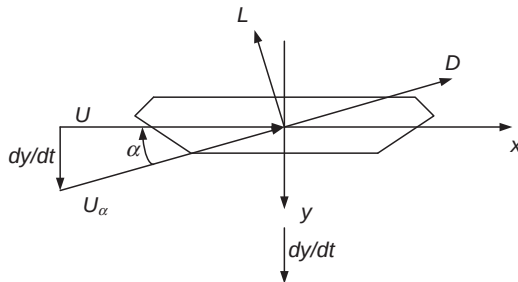


Figure 4.4 Schematic diagram for 2-D steady flow-induced galloping.

Assume that the oscillation is small, thus $\alpha \approx \frac{\dot{y}}{U} \rightarrow 0$. Applying Taylor's expansion to Equation 4.11 and considering only the first two terms yields:

$$F_y(\alpha) = F_y(0) + \left. \frac{\partial F_y}{\partial \alpha} \right|_{\alpha=0} \cdot \alpha = \frac{1}{2} \rho U^2 B C_L + \frac{1}{2} \rho U^2 B \left(\frac{dC_L}{d\alpha} + C_D \right) \bigg|_{\alpha=0} \cdot \frac{\dot{y}}{U} \quad (4.12)$$

Neglecting the static components yields:

$$F_y(\alpha) = \frac{1}{2} \rho U^2 B \left(\frac{dC_L}{d\alpha} + C_D \right) \bigg|_{\alpha=0} \cdot \frac{\dot{y}}{U} \quad (4.13)$$

which is a quasi-steady expression of aerodynamic force acting on the structure section in the y-direction.

4.3.2 Criterion for Galloping Instability

The governing equation of vertical vibration of the SDOF structure can be written as:

$$m\ddot{y} + c\dot{y} + ky = \frac{1}{2} \rho U^2 B \left(\frac{dC_L}{d\alpha} + C_D \right) \bigg|_{\alpha=0} \cdot \frac{\dot{y}}{U} \quad (4.14)$$

where:

c is the damping coefficient of the structure;

k is the stiffness coefficient.

The right side of the above equation represents the aerodynamic damping force. Galloping happens when the negative aerodynamic damping force exceeds the structural damping force. It is obvious that if galloping instability is to occur, at least the following equation should be satisfied:

$$\frac{dC_L}{d\alpha} + C_D < 0 \quad (4.15)$$

The above equation can be used to judge the galloping instability of a flexible structure. Clearly, circular cylinders cannot gallop. This equation is called the Glauert-Hartog criterion. A more exact condition (sufficient condition) of galloping is:

$$\frac{dC_L}{d\alpha} + C_D < -\frac{2c}{\rho U B} \quad (4.16)$$

4.3.3 Wake Galloping

Wake galloping is a phenomenon in which oscillations of a downstream cylinder are induced by the flow in the turbulent wake of an upstream cylinder. Consider two cylinders separated at a few diameters away from each other with one cylinder in the wake of the other, as shown in Figure 4.5. Due to the circulation of the flow inside the wake (clockwise for the upper half of the wake in Figure 4.5 and

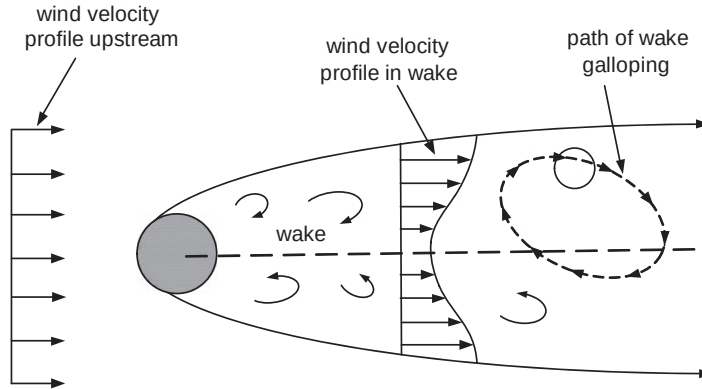


Figure 4.5 Wake galloping (from [3]).

counterclockwise of the lower half), the cylinder located in the upper half of the wake, if allowed to move, will oscillate in a clockwise, elliptic path as shown in Figure 4.5. Likewise, a cylinder free to move in the lower half would oscillate in the counterclockwise direction. Such oscillation or vibration is wake galloping [3].

Wake galloping can happen to stay cables in a cable-stayed bridge, or to the hangers of a suspension bridge. It occurs at high wind speeds and leads to large amplitude oscillations. These oscillations have been found to cause fatigue of the outer strands of hangers at end clamps on suspension bridges, and similar fatigue problems are a theoretical possibility on cable-stayed bridges. Nevertheless, wake galloping occurs only under conditions where the frequency of vibration of the downstream cylinder is low compared to the vortex-shedding frequencies of both the downstream and upstream cylinders. In such cases, when the downstream cylinder is located a few diameters behind the upstream cylinder, it enters a region of galloping instability. Researches show that the instability region is about $8D$ – $20D$, where D is the diameter of the upstream cylinder.

4.4 Flutter Analysis

4.4.1 Introduction

Flutter is a self-feeding and potentially destructive vibration to a long-span cable-supported bridge where aerodynamic forces on the bridge deck couple with its motion. If the energy input by the aerodynamic forces due to strong winds in a cycle is larger than that dissipated by the damping in the bridge structure system, the amplitude of vibration of the bridge deck will increase. This increasing vibration will then amplify the aerodynamic forces, resulting in self-excited forces and self-exciting oscillations. The vibration amplitude of the bridge deck can build up until it results in the collapse of the bridge. One famous example of flutter phenomena was the collapse of the original Tacoma Narrows Bridge in 1940.

The term of classical flutter is originally applied to thin airfoils in the aircraft industry. It means an aeroelastic phenomenon in which two degrees of freedom (DOFs) of a structure, torsional and vertical, couple together in a flow-driven, unstable oscillation. It is also called 2-D flutter. Single degree of freedom (1-D) flutter may manifest itself in a vertical or torsional mode of vibration of a structure, but torsional action is more serious by far. The celebrated failure of the original Tacoma Narrows Bridge exhibited two forms of 1-D flutter – initially a non-catastrophic vertical flutter and then, at a higher wind speed, a large-amplitude torsional flutter [9]. For modern long-span cable-supported bridges, flutter instability may involve multiple modes of vibration.

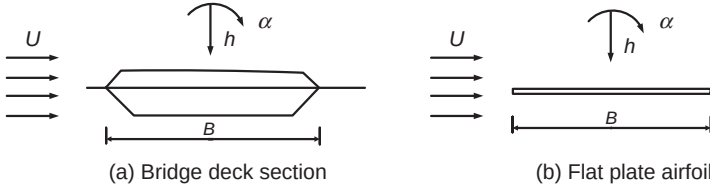


Figure 4.6 2-D structures for flutter analysis.

4.4.2 Self-Excited Forces and Aerodynamic Derivatives

Equation 4.13 gives the self-excited force in a 1-D case, in which only the first two terms are considered in Taylor's expansion and the static component is not considered. In a more general way, by considering only the first two terms in Taylor's expansion and by ignoring the static components, the self-excited forces on a 2-D structure involving both vertical and torsional vibrations (see Figure 4.6a) can be written as:

$$L_{se} = \frac{\partial L_{se}}{\partial \dot{h}} \dot{h} + \frac{\partial L_{se}}{\partial \dot{\alpha}} \dot{\alpha} + \frac{\partial L_{se}}{\partial h} h + \frac{\partial L_{se}}{\partial \alpha} \alpha \quad (4.17a)$$

$$M_{se} = \frac{\partial M_{se}}{\partial \dot{\alpha}} \dot{\alpha} + \frac{\partial M_{se}}{\partial \dot{h}} \dot{h} + \frac{\partial M_{se}}{\partial \alpha} \alpha + \frac{\partial M_{se}}{\partial h} h \quad (4.17b)$$

where:

L_{se} and M_{se} are the self-excited lift and moment on the structure respectively;
 h and α are the vertical and torsional displacements of the structure, respectively.

The self-excited lift and moment are non-linear functions of the vertical and torsional displacements and their derivatives. However, the incipient flutter condition, which separates the stable and unstable regimes, may be treated as having only small amplitude. With such assumptions, Scanlan and Tomko [10] introduced aerodynamic derivatives to express the self-excited forces as:

$$L_{se} = \frac{1}{2} \rho \bar{U}^2 B \left[KH_1^* \frac{\dot{h}}{\bar{U}} + KH_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B} \right] \quad (4.18a)$$

$$M_{se} = \frac{1}{2} \rho \bar{U}^2 B^2 \left[KA_1^* \frac{\dot{h}}{\bar{U}} + KA_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B} \right] \quad (4.18b)$$

where:

$\bar{U}$ is the mean wind velocity;
 H_i^* and A_i^* ($i = 1-4$) are the aerodynamic derivatives.

The aerodynamic derivatives can be obtained from either wind tunnel tests or computational fluid dynamics (CFD) simulation, which will be introduced in Chapters 7 and 8, respectively.

4.4.3 Theodorsen Circulatory Function

The theoretical expressions for self-excited lift and moment on a flat plate airfoil subject to sinusoidal motions (see Figure 4.6b) are given by Theodorsen [11]:

$$L_{se} = \pi \rho b \left\{ -b\ddot{h} - 2UC(k)\dot{h} - [1 + C(k)]Ub\dot{\alpha} - 2U^2C(k)\alpha \right\} \quad (4.19a)$$

$$M_{se} = \pi \rho b^2 \left\{ UC(k)\dot{h} - \frac{b^2\ddot{\alpha}}{8} + \left[-\frac{1}{2} + \frac{1}{2}C(k) \right] Ub\dot{\alpha} + U^2C(k)\alpha \right\} \quad (4.19b)$$

where:

b is the half of the plate width;

$C(k)$ is the Theodorsen cyclical function, which can be expressed as:

$$C(k) = F(k) + iG(k) \quad (4.20)$$

The functions $F(K)$ and $G(K)$ are the real and imaginary parts of the function $C(k)$, as shown in Figure 4.7, which can be expressed mathematically in terms of the Bessel function of the first and second kind.

In consideration of sinusoidal displacements of the flat plate airfoil in Equation 4.19 and then by comparing this equation with Equation 4.18, the aerodynamic derivatives of the flat plate airfoil can be found as:

$$H_1^*(K) = -\frac{\pi F(k)}{k} \quad A_1^*(K) = -\frac{\pi F(k)}{4k} \quad (4.21a)$$

$$H_2^*(K) = \frac{\pi}{4k} \left[1 + F(k) + \frac{2G(k)}{k} \right] \quad A_2^*(K) = -\frac{\pi}{16k} \left[1 - F(k) - \frac{2G(k)}{k} \right] \quad (4.21b)$$

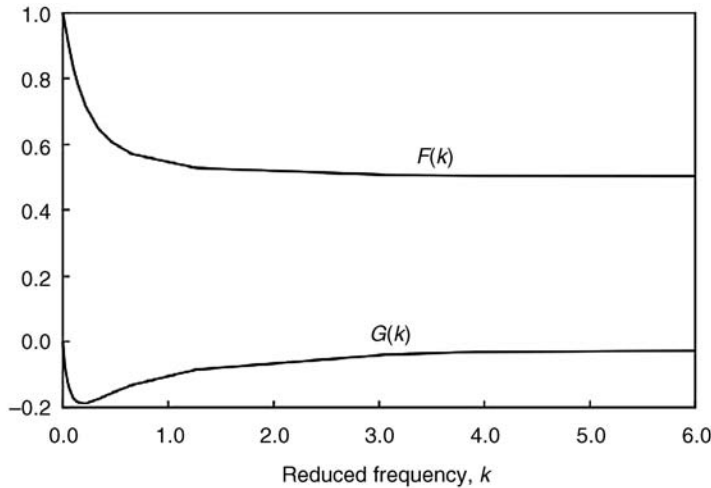


Figure 4.7 Real and imaginary parts of the Theodorsen function.

$$H_3^*(K) = \frac{\pi}{2k^2} \left[F(k) - \frac{kG(k)}{2} \right] \quad A_3^*(K) = -\frac{\pi}{8k^2} \left[F(k) - \frac{kG(k)}{2} \right] \quad (4.21c)$$

$$H_4^*(K) = \frac{\pi}{2} \left[1 + \frac{2G(k)}{k} \right] \quad A_4^*(K) = -\frac{\pi G(k)}{4k} \quad (4.21d)$$

where $k = K/2$.

4.4.4 1-D Flutter Analysis

The 1-D flutter analysis considered here is the pure torsional motion of the bridge deck. The governing equation of motion of the bridge deck for 1-D flutter analysis can be derived from Equation 4.18b as:

$$I(\ddot{\alpha} + 2\xi_{0\alpha}\omega_{0\alpha}\dot{\alpha} + \omega_{0\alpha}^2\alpha) = \frac{1}{2}\rho\bar{U}^2B^2 \left[KA_2^* \frac{B\dot{\alpha}}{U} + K^2A_3^*\alpha \right] \quad (4.22)$$

where:

I represents the mass moment of inertia of the structure;

$\xi_{0\alpha}$ is the structural damping ratio in the torsional vibration;

$\omega_{0\alpha}$ is the structural circular natural frequency in the torsional vibration.

Thus, the total stiffness k_α for the wind-structure system is:

$$k_\alpha = I\omega_{0\alpha}^2 - \frac{1}{2}\rho\bar{U}^2B^2K^2A_3^*(K) \quad (4.23)$$

From the above equation, one can get the vibration frequency for the wind-structure system as:

$$\omega_\alpha = \omega_{\alpha 0} \sqrt{1 - \frac{\rho B^4}{2I} \frac{\omega_\alpha^2}{\omega_{\alpha 0}^2} A_3^* \left(\frac{\omega_\alpha B}{U} \right)} \approx \omega_{\alpha 0} \sqrt{1 - \frac{\rho B^4}{2I} A_3^* \left(\frac{\omega_{\alpha 0} B}{U} \right)} \quad (4.24)$$

In a similar way, the total damping ξ_α for the wind-structure system can be expressed as:

$$\xi_\alpha = 2I\xi_{0\alpha}\omega_{0\alpha} - \frac{1}{2}\rho\bar{U}B^3KA_2^*(K) \quad (4.25)$$

It is clear that if A_2^* is negative, the total damping is positive and the torsional vibration of the structure is stable. If bridge decks have positive A_2^* , the total damping may be negative. This causes flutter to occur and vibration amplitude will increase until the structure collapses. Therefore, the critical condition can be expressed as:

$$2I\xi_{0\alpha}\omega_{0\alpha} = \frac{1}{2}\rho U_{cr} B^3 K A_2^*(K) \quad (4.26)$$

Because $K = \omega_\alpha B / \bar{U}$, the above equation can be rewritten as:

$$A_2^* \left(\frac{\omega_\alpha B}{U_{cr}} \right) = \frac{4I}{\rho B^4} \xi_{0\alpha} \frac{\omega_{0\alpha}}{\omega_\alpha} \quad (4.27)$$

Using Equations 4.27 and 4.24, the critical flutter wind speed for pure torsional motion can be calculated directly.

4.4.5 2-D Flutter Analysis

2-D flutter analysis aims to calculate the critical flutter wind speed for an ideal structure with two degrees of freedom – normally, one in the vertical direction and the other in the torsional direction.

Although a long-span cable-supported bridge is a complicated structure in which flutter may involve multiple modes of vibration, the first vertical mode of vibration and the first torsional mode of vibration are often considered to be most important. Moreover, in terms of the modal decomposition method, the first vertical mode and the first torsional mode can be simplified into a 2-D system. Therefore, a simplified 2-D flutter analysis can be performed for a real bridge by assuming that only the first vertical and torsional modes participate in the coupled flutter, although the critical wind speed derived from such a flutter analysis may be over-estimated due to the simplification.

It can be seen from Equation 4.18a that the self-excited forces are coupled in the vertical and torsional directions. Due to this coupling effect, a bridge can have flutter instability even if its girder section has a negative aerodynamic damping derivative A_2^* in the torsional direction.

The governing equation of motion of the structure for 2-D flutter analysis can be written as:

$$M(\ddot{h} + 2\xi_{0h}\omega_{0h}\dot{h} + \omega_{0h}^2 h) = \frac{1}{2}\rho\bar{U}^2 B \left[KH_1^* \frac{\dot{h}}{\bar{U}} + KH_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B} \right] \quad (4.28a)$$

$$I(\ddot{\alpha} + 2\xi_{0\alpha}\omega_{0\alpha}\dot{\alpha} + \omega_{0\alpha}^2 \alpha) = \frac{1}{2}\rho\bar{U}^2 B^2 \left[KA_1^* \frac{\dot{h}}{\bar{U}} + KA_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B} \right] \quad (4.28b)$$

where:

M represents the structural mass;

ξ_{0h} denotes the structural damping ratio in the vertical direction;

ω_{0h} denotes the structural natural frequency in the vertical direction.

Let $X = [h \ \alpha]^T$, for brevity. By considering the structure and air as an integrated system and transposing the terms on the right side of Equation 4.28 to the left side, Equation 4.28 can be rewritten as:

$$\ddot{X} + C^e \dot{X} + K^e X = 0 \quad (4.29)$$

where:

$$C^e = \begin{pmatrix} 2\xi_{0h}\omega_{0h} - \frac{\rho B^2 L}{M} \omega_h H_1^* & \frac{\rho B^3 L}{M} \omega_\alpha H_2^* \\ \frac{\rho B^3 L}{I} \omega_h A_1^* & 2\xi_{0\alpha}\omega_{0\alpha} - \frac{\rho B^4 L}{I} \omega_\alpha A_2^* \end{pmatrix} \quad (4.30a)$$

$$K^e = \begin{pmatrix} \omega_{0h}^2 - \frac{\rho B^2 L}{M} \omega_h^2 H_4^* & \frac{\rho B^3 L}{M} \omega_\alpha^2 H_3^* \\ \frac{\rho B^3 L}{I} \omega_h^2 A_4^* & \omega_{0\alpha}^2 - \frac{\rho B^4 L}{I} \omega_\alpha^2 A_3^* \end{pmatrix} \quad (4.30b)$$

C^e and K^e are the damping and stiffness matrix of the coupled wind-structure system respectively; ω_h and ω_α denote the frequency of the vibration system in the vertical and torsional direction respectively.

Letting $Y = [X \ \dot{X}]^T$, the governing equation can be written in the state space as:

$$\dot{Y} = AY \quad (4.31)$$

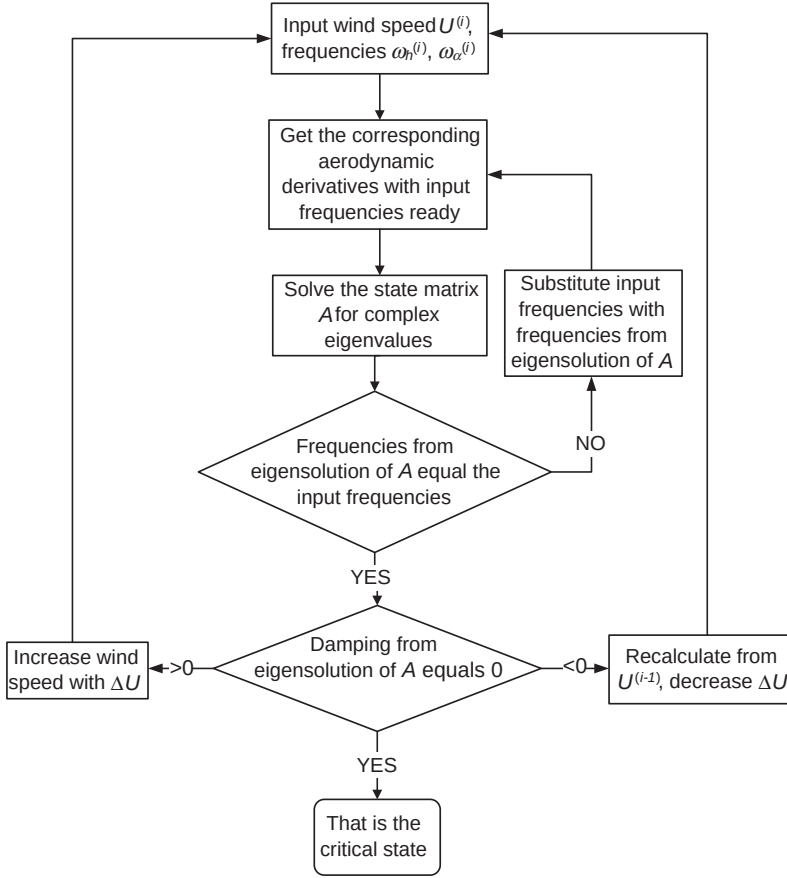


Figure 4.8 Flow chart of 2-D flutter analysis.

where:

$A = \begin{pmatrix} 0 & I \\ -K^e & -C^e \end{pmatrix}$ is the state matrix;

I denotes the unit matrix.

The real parts of the eigenvalues of the matrix A ($-\xi_i \omega_i$, $i = h, \alpha$) represent the modal damping of the coupled system, while the imaginary parts of the eigenvalues of A ($\sqrt{1 - \xi_i^2} \omega_i$, $i = h, \alpha$) represent the modal frequencies of the coupled system. Through an iterative eigenvalue analysis of A , the critical flutter wind speed for a 2-D coupled system can be calculated. The procedure of a 2-D flutter analysis is shown in Figure 4.8.

4.4.6 3-D Flutter Analysis in the Frequency Domain

The main purpose of a flutter analysis is to determine the critical flutter wind speed for a specific bridge. It is usually conducted in the frequency domain for calculation efficiency. The theoretical

backgrounds of coupled flutter analysis of long-span cable-supported bridges can be found in the literature [12–16] and are briefly introduced in this section. It is assumed that the buffeting forces have no influence on aerodynamic stability and are excluded in the flutter analysis. As a result, the governing equation of motion of a bridge deck under self-excited forces is given by:

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} = \mathbf{F}_{se} \quad (4.32)$$

where:

$\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the structural mass, damping, and stiffness matrix, respectively;
 $\mathbf{X}$, $\dot{\mathbf{X}}$, $\ddot{\mathbf{X}}$ are the nodal displacement, velocity, and acceleration vector, respectively;
 $\mathbf{F}$ indicates the nodal equivalent force vector; the subscript *se* represents the self-excited force.

The self-excited vertical and lateral forces and self-excited moment acting on the bridge deck per unit length are expressed often in terms of Scanlan's format as follows:

$$L_{se}(t) = \frac{1}{2} \rho \bar{U}^2 (2B) \left(KH_1^* \frac{\dot{h}}{\bar{U}} + KH_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B} + KH_5^* \frac{\dot{p}}{\bar{U}} + K^2 H_6^* \frac{p}{B} \right) \quad (4.33a)$$

$$D_{se}(t) = \frac{1}{2} \rho \bar{U}^2 (2B) \left(KP_1^* \frac{\dot{p}}{\bar{U}} + KP_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 P_3^* \alpha + K^2 H_4^* \frac{h}{B} + KH_5^* \frac{\dot{p}}{\bar{U}} + K^2 H_6^* \frac{p}{B} \right) \quad (4.33b)$$

$$M_{se}(t) = \frac{1}{2} \rho \bar{U}^2 (2B^2) \left(KA_1^* \frac{\dot{h}}{\bar{U}} + KA_2^* \frac{B\dot{\alpha}}{\bar{U}} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B} + KA_5^* \frac{\dot{p}}{\bar{U}} + K^2 A_6^* \frac{p}{B} \right) \quad (4.33c)$$

where:

h , p , and α are the vertical, lateral, and torsional displacements of the bridge deck, respectively;
the over-dot denotes the partial differentiation with respect to time t ;
 H_i^* , P_i^* , A_i^* ($i = 1 - 6$) are the non-dimensional aerodynamic derivatives (flutter derivatives) which can be obtained from wind tunnel tests.

At present, a two-degrees-of-freedom section model of the bridge deck is widely used to identify the flutter derivatives H_i^* and A_i^* ($i = 1 - 4$). The drag components associated with the lateral motion and some coupling terms are generally negligible. If these flutter derivatives are not available from the wind tunnel tests, the empirical expressions based on the quasi-steady theory can be used in the analysis:

$$P_1^* = -\frac{1}{K} C_D, \quad P_2^* = \frac{1}{2K} C'_D, \quad P_3^* = \frac{1}{2K^2} C'_D \quad (4.34a)$$

$$P_5^* = \frac{1}{2K} C'_D, \quad H_5^* = \frac{1}{K} C_L, \quad A_5^* = -\frac{1}{K} C_M \quad (4.34b)$$

$$P_4^* = P_6^* = H_6^* = A_6^* = 0 \quad (4.34c)$$

where:

C_L , C_D , and C_M are the aerodynamic lift, drag, and moment coefficients referring to the deck width B , respectively;
 $C'_D = dC_D/d\alpha$.

By using a complex notation, the self-excited forces on the bridge deck per unit length may be expressed as:

$$L_{se}(t) = \omega^2 \rho B^2 (C_{Lh}h + C_{Lp}p + BC_{L\alpha}\alpha) \quad (4.35a)$$

$$D_{se}(t) = \omega^2 \rho B^2 (C_{Dh}h + C_{Dp}p + BC_{D\alpha}\alpha) \quad (4.35b)$$

$$M_{se}(t) = \omega^2 \rho B^2 (BC_{Mh}h + BC_{Mp}p + B^2C_{M\alpha}\alpha) \quad (4.35c)$$

where C_{rs} ($r = D, L, M$; $s = h, p, \alpha$) are the complex flutter derivatives of self-excited forces.

The relationships between the real and complex flutter derivatives can be found as:

$$C_{Lh} = H_4^* + iH_1^*, \quad C_{Lp} = H_6^* + iH_5^*, \quad C_{L\alpha} = H_3^* + iH_2^* \quad (4.36a)$$

$$C_{Dh} = P_6^* + iP_5^*, \quad C_{Dp} = P_4^* + iP_1^*, \quad C_{D\alpha} = P_3^* + iP_2^* \quad (4.36b)$$

$$C_{Mh} = A_4^* + iA_1^*, \quad C_{Mp} = A_6^* + iA_5^*, \quad C_{M\alpha} = A_3^* + iA_2^* \quad (4.36c)$$

In a 3-D finite element-based flutter analysis, the distributed self-excited forces acting on an element of a bridge deck need to be converted into equivalent nodal loads at two ends of the element.

$$\mathbf{F}_{se}^e = \omega^2 \mathbf{A}_{se}^e \mathbf{X}^e \quad (4.37)$$

where:

the subscript e represents the local coordinates of the element;

$\mathbf{A}_{se}^e$ is a 12×12 aeroelastic matrix.

For an element with length L , the aeroelastic matrix is

$$\mathbf{A}_{se}^e = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_1 \end{bmatrix} \quad (4.38)$$

where:

$$\mathbf{A}_1 = \frac{1}{2} \rho B^2 L \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & C_{Lh} & C_{Lp} & BC_{L\alpha} & 0 & 0 \\ 0 & C_{Dh} & C_{Dp} & BC_{D\alpha} & 0 & 0 \\ 0 & BC_{Mh} & BC_{Mp} & B^2C_{M\alpha} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (4.39)$$

Since aeroelastic (self-excited) forces are non-conservative, the aeroelastic matrix of the element is generally unsymmetric and it is a function of the reduced frequency. When the aeroelastic matrix of the element is transformed into the global coordinate system and is assembled, one may have:

$$\mathbf{F}_{se} = \omega^2 \mathbf{A}_{se} \mathbf{X} \quad (4.40)$$

where $\mathbf{A}_{se}$ is the aeroelastic matrix of the bridge, and it is a complex matrix. The governing equation of motion of the bridge for flutter analysis can then be expressed as:

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} = \omega^2 \mathbf{A}_{se}\mathbf{X} \quad (4.41)$$

Let $\mathbf{X}$ equal $\mathbf{R} e^{st}$, where $\mathbf{R}$ is the complex modal response amplitude vector of the system. Denote the complex frequency $s = (-\xi + i)\omega$ (where ξ and ω are the damping ratio and circular frequency of the complex mode of vibration, respectively, and $i^2 = -1$). The governing equation of motion can be written as:

$$(s^2 \mathbf{M} + s \mathbf{C} + \mathbf{K} - \omega^2 \mathbf{A}_{se}) \mathbf{R} e^{st} = \mathbf{0} \quad (4.42)$$

The complex mode response of the system can be given approximately by the first m modes of vibration:

$$\mathbf{R} = \Phi \mathbf{q} \quad (4.43)$$

where:

Φ is the matrix of mode shapes, obtained from the modal analysis of the structure;
 $\mathbf{q}$ is the vector of generalized coordinates.

Applying the above linear transformation to Equation 4.41 yields:

$$[s^2 \mathbf{I} - \omega^2 \bar{\mathbf{A}}_{se} + s \bar{\mathbf{C}} + \Lambda] \mathbf{q} e^{st} = \mathbf{0} \quad (4.44)$$

where Λ is the diagonal eigenvalue matrix obtained from modal analysis; and the matrix $\bar{\mathbf{A}}_{se} = \Phi^T \mathbf{A}_{se} \Phi$ and $\bar{\mathbf{C}} = \Phi^T \mathbf{C} \Phi$.

Considering the fact that the damping ratios of the system (positive or negative) are small, the approximate relation $\omega^2 = -s^2$ exists. As a result, one may have:

$$[s^2 (\mathbf{I} + \bar{\mathbf{A}}_{se}) + s \bar{\mathbf{C}} + \Lambda] \mathbf{q} e^{st} = \mathbf{0} \quad (4.45)$$

The above equation can be further expressed in the following state-space format:

$$(\mathbf{A} - s \mathbf{I}) \mathbf{Y} e^{st} = \mathbf{0} \quad (4.46)$$

where:

$$\mathbf{Y} = \begin{Bmatrix} \mathbf{q} \\ s \mathbf{q} \end{Bmatrix}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\bar{\mathbf{M}} \Lambda & -\bar{\mathbf{M}} \bar{\mathbf{C}} \end{bmatrix} \quad (4.47a)$$

$$\bar{\mathbf{M}} = (\mathbf{I} + \bar{\mathbf{A}}_{se})^{-1} \quad (4.47b)$$

Thus, to have a non-trivial solution, the following equation must be satisfied, leading to a standard eigenvalue problem:

$$\mathbf{A} \mathbf{Y} = s \mathbf{Y} \quad (4.48)$$

where the characteristic matrix $\mathbf{A}$ is a $2m \times 2m$ complex matrix and a function of reduced frequency K (or reduced velocity) only. Thus, the above equation can be solved for only two variables, s and K .

For a given K , standard linear eigensolvers are available to find the $2m$ sets of eigenvalues s and corresponding eigenvectors $\mathbf{Y}$ from Equation 4.48:

$$s = (-\xi + i)\omega \quad (4.49a)$$

$$\mathbf{q} = \mathbf{a} + \mathbf{b}i \quad (4.49b)$$

The m eigenvalues with positive imaginary part are the complex frequencies of the system, and the upper half vector $\mathbf{q}$ in the corresponding eigenvector $\mathbf{Y}$ is the complex mode shape of the system. In a prescribed complex mode shape, the magnitude and phase of the k th natural mode are given as:

$$|q_k| = \sqrt{a_k^2 + b_k^2} \quad (4.50a)$$

$$\phi_k = \tan^{-1}(b_k/a_k) \quad (4.50b)$$

If the damping ratios of all complex modes are positive, the system is stable; if at least one damping ratio is equal to zero, the system is neutrally stable; if at least one damping ratio is negative, the system is unstable. Therefore, the flutter analysis described above is able to find the critical state through searching the reduced frequency K . The corresponding circular frequency is the flutter circular frequency ω_f and the critical wind speed U_{cr} is then equal to $B\omega_f/K$. At the critical wind speed, the generalized modal coordinate vector $\mathbf{q}(t)$ and the nodal displacement vector of the bridge can be expressed as:

$$\mathbf{q}(t) = \{|q_i| \sin(\omega_f t + \phi_i)\} \quad (4.51)$$

$$\mathbf{X}(t) = \sum_{i=1}^m \boldsymbol{\varphi}_i |q_i| \sin(\omega_f t + \phi_i) = \mathbf{X}_0 \sin(\omega_f t + \phi_i) \quad (4.52)$$

where:

$\boldsymbol{\varphi}_i$ is the i th natural mode shape;

ω_f is the flutter circular frequency;

X_0 and ϕ_i are the amplitude and phase of $X(t)$;

m is the number of participating modes.

It is clear that the coupled flutter motion is three-dimensional and that the phase shift exists among mode components. The total energy in the characteristic motion (flutter motion) of the bridge at the lowest critical wind speed is:

$$E = \frac{1}{2} \{\dot{\mathbf{X}}_{\max}\}^T \mathbf{M} \{\dot{\mathbf{X}}_{\max}\} = \frac{1}{2} \omega_f^2 \sum_{i=1}^m |q_i|^2 \quad (4.53)$$

The energy in the i th natural mode of vibration of the bridge is expressed as:

$$E_i = \frac{1}{2} \omega_f^2 |q_i|^2 \quad (4.54)$$

The ratio of the i th modal energy over the total energy E_i/E is defined as the modal energy ratio e_i . Clearly, the modal energy ratio provides a uniform measurement of the contribution of a particular vibration mode to the flutter instability of the whole bridge.

4.4.7 Flutter Analysis in the Time Domain

Flutter analysis has been predominantly conducted in the frequency domain for computational efficiency particularly because self-excited forces are functions of reduced frequency.

However, the frequency domain approach is restricted to linear structures excited by the stationary wind loads without considering aerodynamic and/or structural non-linearities. Schemes for time-domain coupled multimode flutter analysis have been proposed by introducing the unsteady self-excited forces in terms of rational function approximations [17]. The advantages of the time-domain approach are that an iterative solution for determining flutter conditions is unnecessary because the equations of motion are independent of frequency, and aerodynamic non-linearities can be taken into account in flutter analysis. The time-domain modeling of self-excited forces is also used in the time-domain buffeting analysis, which will be introduced in Section 4.7 in detail.

4.5 Buffeting Analysis in the Frequency Domain

4.5.1 Background

When a long-span cable-supported bridge is immersed in a wind field, the bridge will be subjected to static and dynamic wind forces caused by mean and fluctuating wind speeds, respectively. Buffeting action on a long-span bridge is a random vibration caused by fluctuating winds that appear within a wide range of wind speeds. In wind resistance design of a long-span bridge, consideration of the buffeting responses is normally dominant in order to determine the size of structural members. In addition to buffeting action, the self-excited forces induced by wind-structure interaction is also important for predicting the buffeting response of long-span bridges, as the additional energy injected into the oscillating structure by self-excited forces increases the magnitude of vibrations. To model the action of buffeting wind load, the buffeting forces resulting from turbulent wind and the self-excited forces due to the wind-bridge interaction should be taken into account. The buffeting response prediction can be performed in both the frequency domain [18–21] and the time domain [17,22,23].

4.5.2 Buffeting Forces and Aerodynamic Admittances

Under the quasi-steady assumption, the transient aerodynamic forces on the bridge deck per unit length can be written as:

$$L_\alpha(t) = \frac{1}{2} \rho \bar{U}^2(t) C_L(\alpha_0 + \Delta\alpha) \cdot B \quad (4.55a)$$

$$D_\alpha(t) = \frac{1}{2} \rho \bar{U}^2(t) C_D(\alpha_0 + \Delta\alpha) \cdot B \quad (4.55b)$$

$$M_\alpha(t) = \frac{1}{2} \rho \bar{U}^2(t) C_M(\alpha_0 + \Delta\alpha) \cdot B^2 \quad (4.55c)$$

where:

the subscript α denotes that the force is expressed in the transient wind axis;

α_0 is the attack angle of mean wind speed $\bar{U}$;

$\Delta\alpha$ is the additional attack angle induced by fluctuation (see Figure 4.9).

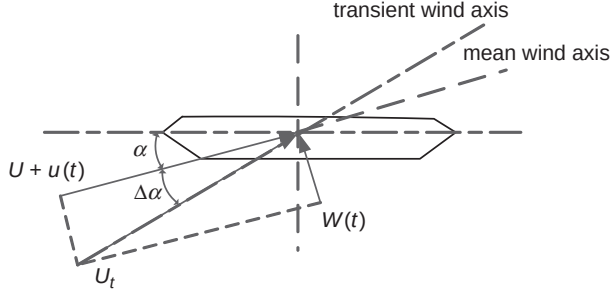


Figure 4.9 Wind and buffeting forces on bridge deck.

The aerodynamic coefficients can be expressed in terms of the Taylor's expansion to the first two terms:

$$C_L(\alpha_0 + \Delta\alpha) = C_L(\alpha_0) + C'_L(\alpha_0) \cdot \Delta\alpha \quad (4.56a)$$

$$C_D(\alpha_0 + \Delta\alpha) = C_D(\alpha_0) + C'_D(\alpha_0) \cdot \Delta\alpha \quad (4.56b)$$

$$C_M(\alpha_0 + \Delta\alpha) = C_M(\alpha_0) + C'_M(\alpha_0) \cdot \Delta\alpha \quad (4.56c)$$

Equation 4.55a can be transformed into the mean wind speed axis as:

$$L(t) = L_\alpha(t) \cdot \cos(\Delta\alpha) + D_\alpha(t) \cdot \sin(\Delta\alpha) \quad (4.57a)$$

$$D(t) = D_\alpha(t) \cdot \cos(\Delta\alpha) - L_\alpha(t) \cdot \sin(\Delta\alpha) \quad (4.57b)$$

$$M(t) = M_\alpha(t) \quad (4.57c)$$

Assuming that $\Delta\alpha$ is very small, then $\sin(\Delta\alpha) \approx \alpha$ and $\cos(\Delta\alpha) \approx 1 - \Delta\alpha^2/2$. Equation 4.57 becomes:

$$L(t) = \frac{1}{2} \rho \bar{U}^2 B \left[C_L(\alpha_0) \cdot \left(\frac{2u(t)}{\bar{U}} \right) + (C'_L(\alpha_0) + C_D(\alpha_0)) \cdot \frac{w(t)}{\bar{U}} \right] + \frac{1}{2} \rho \bar{U}^2 B C_L(\alpha_0) \quad (4.58a)$$

$$D(t) = \frac{1}{2} \rho \bar{U}^2 B \left[C_D(\alpha_0) \cdot \left(\frac{2u(t)}{\bar{U}} \right) + C'_D(\alpha_0) \cdot \frac{w(t)}{\bar{U}} \right] + \frac{1}{2} \rho \bar{U}^2 B C_D(\alpha_0) \quad (4.58b)$$

$$M(t) = \frac{1}{2} \rho \bar{U}^2 B \left[C_M(\alpha_0) \cdot \left(\frac{2u(t)}{\bar{U}} \right) + C'_M(\alpha_0) \cdot \frac{w(t)}{\bar{U}} \right] + \frac{1}{2} \rho \bar{U}^2 B^2 C_M(\alpha_0) \quad (4.58c)$$

It is obvious that the second terms on the right side of the above equation are static forces. Therefore, the buffeting forces can be written as:

$$L_b(t) = \frac{1}{2} \rho \bar{U}^2 B \left[C_L(\alpha_0) \cdot \left(\frac{2u(t)}{\bar{U}} \right) + (C'_L(\alpha_0) + C_D(\alpha_0)) \cdot \frac{w(t)}{\bar{U}} \right] \quad (4.59a)$$

$$D_b(t) = \frac{1}{2} \rho U^2 B \left[C_D(\alpha_0) \cdot \left(\frac{2u(t)}{\bar{U}} \right) + C'_D(\alpha_0) \cdot \frac{w(t)}{\bar{U}} \right] \quad (4.59b)$$

$$M_b(t) = \frac{1}{2} \rho U^2 B \left[C_M(\alpha_0) \cdot \left(\frac{2u(t)}{\bar{U}} \right) + C'_M(\alpha_0) \cdot \frac{w(t)}{\bar{U}} \right] \quad (4.59c)$$

The quasi-steady buffeting theory assumes that the wind loads are fully correlated along the bridge span. Such an assumption can only be satisfied when the turbulence has a larger scale than the bridge width, otherwise the incomplete span-wise correlation should be taken into account when modeling buffeting forces. Davenport used six aerodynamic admittance functions to represent the span-wise correlation of wind loads [18]. After the modification, the buffeting forces can be written as:

$$L_b = \frac{1}{2} \rho \bar{U}^2 B \left[2C_L \chi_{Lu} \frac{u}{\bar{U}} + (C'_L + C_D) \chi_{Lw} \frac{w}{\bar{U}} \right] \quad (4.60a)$$

$$D_b = \frac{1}{2} \rho \bar{U}^2 B \left[2C_D \chi_{Du} \frac{u}{\bar{U}} + C'_D \chi_{Dw} \frac{w}{\bar{U}} \right] \quad (4.60b)$$

$$M_b = \frac{1}{2} \rho \bar{U}^2 B^2 \left[2C_M \chi_{Mu} \frac{u}{\bar{U}} + C'_M \chi_{Mw} \frac{w}{\bar{U}} \right] \quad (4.60c)$$

where:

$C'_L = dC_L/d\alpha$, $C'_D = dC_D/d\alpha$, and $C'_M = dC_M/d\alpha$;

χ_{Lu} , χ_{Lw} , χ_{Du} , χ_{Dw} , χ_{Mu} , χ_{Mw} are the aerodynamic admittance functions, which are functions of the reduced frequency and dependent on the geometrical configuration of the cross section of the bridge deck;

u and w are the longitudinal and vertical turbulences, respectively.

How to determine the aerodynamic admittance functions is a key issue in buffeting analysis. One conventional way is to compare measured wind spectra and buffeting force spectra and then obtain the empirical aerodynamic admittance functions by the following equations:

$$\chi_{Lu}(K) = \frac{4}{\rho^2 \bar{U}^2 B^2} \cdot \frac{S_{L_b L_b}(x, K)}{S_{uu}(x, K)} \quad \chi_{Lw}(K) = \frac{4}{\rho^2 \bar{U}^2 B^2} \cdot \frac{S_{L_b L_b}(x, K)}{S_{ww}(x, K)} \quad (4.61a)$$

$$\chi_{Du}(K) = \frac{4}{\rho^2 \bar{U}^2 B^2} \cdot \frac{S_{D_b D_b}(x, K)}{S_{uu}(x, K)} \quad \chi_{Dw}(K) = \frac{4}{\rho^2 \bar{U}^2 B^2} \cdot \frac{S_{D_b D_b}(x, K)}{S_{ww}(x, K)} \quad (4.61b)$$

$$\chi_{Mu}(K) = \frac{4}{\rho^2 \bar{U}^2 B^4} \cdot \frac{S_{M_b M_b}(x, K)}{S_{uu}(x, K)} \quad \chi_{Mw}(K) = \frac{4}{\rho^2 \bar{U}^2 B^4} \cdot \frac{S_{M_b M_b}(x, K)}{S_{ww}(x, K)} \quad (4.61c)$$

where:

$S_{i_b i_b}(x, K)$ is the buffeting force spectrum ($i = L, D$, and M);

$S_{jj}(x, K)$ is the wind spectrum ($j = u, w$).

4.5.3 3-D Buffeting Analysis in the Frequency Domain

The main purpose of buffeting analysis is to calculate the dynamic response of a bridge under both self-excited and buffeting forces. It can be conducted in either frequency domain or time domain. This section will introduce buffeting analysis in the frequency domain, which offers better computational efficiency because the analysis can be carried in the selected mode. The governing equation of motion of a bridge excited by fluctuating winds with respect to the static equilibrium position can be given in a matrix form by:

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} = \mathbf{F}_{se} + \mathbf{F}_b + \mathbf{F}_s \quad (4.62)$$

where:

$\mathbf{F}$ indicates the nodal equivalent force vector;

the subscript se , b and s represents the self-excited force, buffeting force and mean wind force components, respectively.

The mean deformation of the bridge caused by the mean wind forces can be readily determined using the mean wind speeds and the other parameters determined by wind tunnel tests as introduced in Chapter 3. The formulation of self-excited force $\mathbf{F}_{se}$ is already mentioned in the above section, as expressed by Equation 4.40. The vertical and lateral buffeting forces and buffeting moment acting on the bridge deck per unit length due to wind fluctuations are given in Equation 4.60. If the aerodynamic admittance functions are taken as units, the buffeting forces aforementioned can be expressed as follows:

$$\mathbf{F}_b = 0.5\rho\overline{U}(\mathbf{C}_{bu}u + \mathbf{C}_{bw}w) \quad (4.63)$$

where:

$$\mathbf{F}_b = \begin{Bmatrix} L_b \\ D_b \\ M_b \end{Bmatrix}, \quad \mathbf{C}_{bu} = B \begin{Bmatrix} 2C_L \\ 2C_D \\ 2BC_M \end{Bmatrix}, \quad \mathbf{C}_{bw} = B \begin{Bmatrix} C'_L + C_D \\ C'_D \\ BC'_M \end{Bmatrix} \quad (4.64)$$

When the element is small enough, it can be assumed that the longitudinal and vertical wind fluctuations are distributed linearly on the element:

$$u = \begin{bmatrix} 1 - \frac{x}{L} & \frac{x}{L} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \mathbf{A}\mathbf{u}^e \quad (4.65a)$$

$$w = \begin{bmatrix} 1 - \frac{x}{L} & \frac{x}{L} \end{bmatrix} \begin{Bmatrix} w_1 \\ w_2 \end{Bmatrix} = \mathbf{A}\mathbf{w}^e \quad (4.65b)$$

where:

x and L are the axial location and the length of the element, respectively;
the subscripts 1 and 2 indicate the two ends of the element.

The consistent buffeting forces at the element ends in the local coordinate system can be obtained by the following definite integral:

$$\mathbf{F}_b^e = \int_L \mathbf{B}^T \mathbf{F}_b dx = 0.5\rho\overline{U} \left(\int_L \mathbf{B}^T \mathbf{C}_{bu} \mathbf{A} dx \mathbf{u}^e + \int_L \mathbf{B}^T \mathbf{C}_{bw} \mathbf{A} dx \mathbf{w}^e \right) = 0.5\rho\overline{U} (\mathbf{A}_{bu}^e \mathbf{u}^e + \mathbf{A}_{bw}^e \mathbf{w}^e) \quad (4.66)$$

where:

$\mathbf{A}_{bu}^e$ and $\mathbf{A}_{bw}^e$ are the buffeting force matrices of the element corresponding to the longitudinal and vertical wind fluctuations, respectively;

B is the matrix of interpolated functions:

$$B = \begin{bmatrix} 0 & -N_1 & 0 & 0 & 0 & -N_3 & 0 & -N_2 & 0 & 0 & 0 & N_4 \\ 0 & 0 & -N_1 & 0 & N_3 & 0 & 0 & 0 & -N_2 & 0 & -N_4 & 0 \\ 0 & 0 & 0 & -N_5 & 0 & 0 & 0 & 0 & 0 & -N_6 & 0 & 0 \end{bmatrix} \quad (4.67)$$

where:

$$N_1 = 1 - 3\left(\frac{x}{L}\right)^2 + 2\left(\frac{x}{L}\right)^3; \quad N_2 = 3\left(\frac{x}{L}\right)^2 - 2\left(\frac{x}{L}\right)^3; \quad N_3 = x\left(1 - \frac{x}{L}\right)^2; \quad (4.68a)$$

$$N_4 = \frac{x^2}{L}\left(1 - \frac{x}{L}\right); \quad N_5 = 1 - \frac{x}{L}; \quad N_6 = \frac{x}{L} \quad (4.68b)$$

The matrices $\mathbf{A}_{bu}^e$ and $\mathbf{A}_{bw}^e$ can be derived as:

$$\mathbf{A}_{bu}^e = \frac{-BL}{30} \begin{bmatrix} 0 & 21C_L & 21C_D & 20BC_M & -3LC_D & 3LC_L \\ 0 & 9C_L & 9C_D & 10BC_M & -2LC_D & 2LC_L \\ 0 & 9C_L & 9C_D & 10BC_M & 2LC_D & -2LC_L \\ 0 & 21C_L & 21C_D & 20BC_M & 3LC_D & -3LC_L \end{bmatrix}^T \quad (4.69)$$

$$\mathbf{A}_{bw}^e = \frac{-BL}{60} \begin{bmatrix} 0 & 21(C'_L + C_D) & 21C'_D & 20BC'_M & -3LC'_D & 3L(C'_L + C_D) \\ 0 & 9(C'_L + C_D) & 9C'_D & 10BC'_M & -2LC'_D & 2L(C'_L + C_D) \\ 0 & 9(C'_L + C_D) & 9C'_D & 10BC'_M & 2LC'_D & -2L(C'_L + C_D) \\ 0 & 21(C'_L + C_D) & 21C'_D & 20BC'_M & 3LC'_D & -3L(C'_L + C_D) \end{bmatrix}^T \quad (4.70)$$

The local nodal buffeting forces can be converted into the global coordinate system using the coordinate transformation matrix. As a result, the global nodal buffeting force vector can be obtained as:

$$\mathbf{F}_b = 0.5\rho\bar{U}(\mathbf{A}_{bu}\mathbf{u} + \mathbf{A}_{bw}\mathbf{w}) \quad (4.71)$$

where:

$\mathbf{A}_{bu}$ and $\mathbf{A}_{bw}$ are the global buffeting force matrices;

$\mathbf{u}$ and $\mathbf{w}$ are the r -row nodal fluctuating wind vectors for the longitudinal and vertical components, respectively, where r is the number of nodes subjected to wind fluctuations.

Apart from the bridge deck, the buffeting forces also act on the bridge towers, the cables and other components. These forces can be determined using a similar way to the determination of the forces acting on the bridge deck. It is thus possible to have a buffeting analysis of the bridge as a whole, rather than the bridge deck only. Based on the preceding discussion, the governing equation of motion of the bridge as a whole can be written as:

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} - \omega^2\mathbf{A}_{se}\mathbf{X} = \mathbf{F}_b \quad (4.72)$$

The buffeting response of the bridge is dominant by the first m modes of vibration and thus a linear transformation is introduced as:

$$\mathbf{X} = \Phi\mathbf{q} \quad (4.73)$$

where:

Φ is the matrix of mode shapes, which are obtained from the modal analysis;
 $\mathbf{q}$ is the m vector of generalized coordinates.

The equation of motion of the bridge can be then expressed as:

$$\ddot{\mathbf{q}} + \overline{\mathbf{C}}\dot{\mathbf{q}} + \Lambda\mathbf{q} - \omega^2\overline{\mathbf{A}}_{se}\mathbf{q} = \mathbf{Q}_b \quad (4.74)$$

where:

Λ is the diagonal eigenvalue matrix obtained from the dynamic characteristic analysis;
 $\overline{\mathbf{C}} = \Phi^T \mathbf{C} \Phi$; $\overline{\mathbf{A}}_{se} = \Phi^T \mathbf{A}_{se} \Phi$; $\mathbf{Q}_b = \overline{\mathbf{A}}_{bu} \mathbf{u} + \overline{\mathbf{A}}_{bw} \mathbf{w}$ ($\overline{\mathbf{A}}_{bu} = \Phi^T \mathbf{A}_{bu}$ and $\overline{\mathbf{A}}_{bw} = \Phi^T \mathbf{A}_{bw}$) are the generalized buffeting force vector.

According to the random vibration theory, the power spectral density (PSD) matrices of the vectors of generalized modal response $\mathbf{q}$ and nodal displacement $\mathbf{X}$ have the following relations:

$$\mathbf{S}_q(\omega) = \mathbf{H}^*(\omega) \mathbf{S}_{Q_b}(\omega) \mathbf{H}^T(\omega) \quad (4.75)$$

$$\mathbf{S}_X(\omega) = \Phi \mathbf{H}^*(\omega) \mathbf{S}_{Q_b}(\omega) \mathbf{H}^T(\omega) \Phi^T \quad (4.76)$$

where $\mathbf{H}(\omega)$ is the transfer function matrix:

$$\mathbf{H}(\omega) = [-\omega^2(\mathbf{I} + \overline{\mathbf{A}}_{se}) + i\omega\overline{\mathbf{C}} + \Lambda]^{-1} \quad (4.77)$$

in which the subscript $*$ and T denote the complex conjugate and transpose, respectively.

The PSD matrix of the generalized buffeting forces is given by:

$$\mathbf{S}_{Q_b}(\omega) = 0.25\rho^2\overline{U}^2(\overline{\mathbf{A}}_{bu}\mathbf{S}_{uu}\overline{\mathbf{A}}_{bu}^T + \overline{\mathbf{A}}_{bw}\mathbf{S}_{ww}\overline{\mathbf{A}}_{bw}^T) \quad (4.78)$$

where $\mathbf{S}_{uu}$ and $\mathbf{S}_{ww}$ are the PSD matrices of $\mathbf{u}$ and $\mathbf{w}$ components, respectively.

If the aerodynamic admittance functions are taken into consideration, the above PSD matrix includes the aerodynamic admittance functions. The power spectra of wind components $\mathbf{u}$ and $\mathbf{w}$ are functions of the circular frequency ω . The cross-spectral density functions of the wind component between two points can be expressed in a conventional form [24]:

$$S_{uu(ww)}(z_1, z_2, \omega) = \sqrt{S_{uu(ww)}(z_1, \omega)S_{uu(ww)}(z_2, \omega)}e^{-\hat{f}_{u(w)}} \quad (4.79)$$

where:

z_1 and z_2 denote the two points;

$S_{uu(ww)}(z, \omega)$ is the auto spectra identified from the wind characteristics study;

$e^{-\hat{f}_{u(w)}}$ is the coherence function of fluctuating winds given by the wind characteristics study.

The components of the matrices S_q and S_x can be expressed as:

$$S_{q_{ij}}(\omega) = \sum_{k=1}^m \sum_{l=1}^m H_{ik}^*(\omega) S_{qb_{kl}}(\omega) H_{jl}(\omega) \quad (4.80)$$

$$S_{X_i}(\omega) = \sum_{k=1}^m \sum_{l=1}^m \varphi_{ik} S_{q_{kl}}(\omega) \varphi_{il} \quad (4.81)$$

The variances of the i th generalized modal response and nodal displacement are thus given by:

$$\sigma_{q_{ii}}^2 = \int_0^\infty S_{q_{ii}}(\omega) d\omega \quad (4.82)$$

$$\sigma_{X_i}^2 = \int_0^\infty S_{X_i}(\omega) d\omega = \sum_{k=1}^m \sum_{l=1}^m \varphi_{ik} \left(\int_0^\infty S_{q_{kl}}(\omega) d\omega \right) \varphi_{il} \quad (4.83)$$

Figure 4.10 provides the flow chart of 3-D finite element based buffeting analysis for a long-span bridge.

4.6 Simulation of Stationary Wind Field

In order to carry out the buffeting analysis in the time domain, the stationary wind field of turbulent winds should be simulated, and this is generally represented by turbulence wind components $u(t)$, $v(t)$ and $w(t)$. Among various simulation methods, the spectral representation methods appear to be most widely used because they are fast and conceptually straightforward. In this section, the simulation of a stationary wind field mainly concerns the simulation of one-dimensional, multivariate, stationary stochastic process, and the algorithm of the spectral representation method proposed by Shinozuka and Jan [25] and Deodatis [26] is introduced.

Consider a set of n one-dimensional stationary stochastic process $\{u_j^0(t)\} (j=1, 2, \dots, n)$ with zero as the mean value, where the superscript 0 denotes the target function. The two-sided target cross-spectral density function $S^0(\omega)$ of the stochastic process is given by:

$$S^0(\omega) = \begin{bmatrix} S_{11}^0(\omega) & S_{12}^0(\omega) & \cdots & S_{1n}^0(\omega) \\ S_{21}^0(\omega) & S_{22}^0(\omega) & \cdots & S_{2n}^0(\omega) \\ \vdots & \vdots & \ddots & \vdots \\ S_{n1}^0(\omega) & S_{n2}^0(\omega) & \cdots & S_{nn}^0(\omega) \end{bmatrix} \quad (4.84)$$

According to Shinozuka and Jan [25] and Deodatis [26], the stochastic process $\{u_j^0(t)\} (j=1, 2, \dots, n)$ can be simulated by the following series:

$$u_j(t) = 2\sqrt{\Delta\omega} \sum_{m=1}^j \sum_{l=1}^N |H_{jm}(\omega_{ml})| \cos(\omega_{ml}t - \theta_{jm}(\omega_{ml}) + \phi_{ml}) \quad (4.85)$$

where:

N is a sufficiently large number;

$\Delta\omega = \omega_{up}/N$ is the frequency increment;

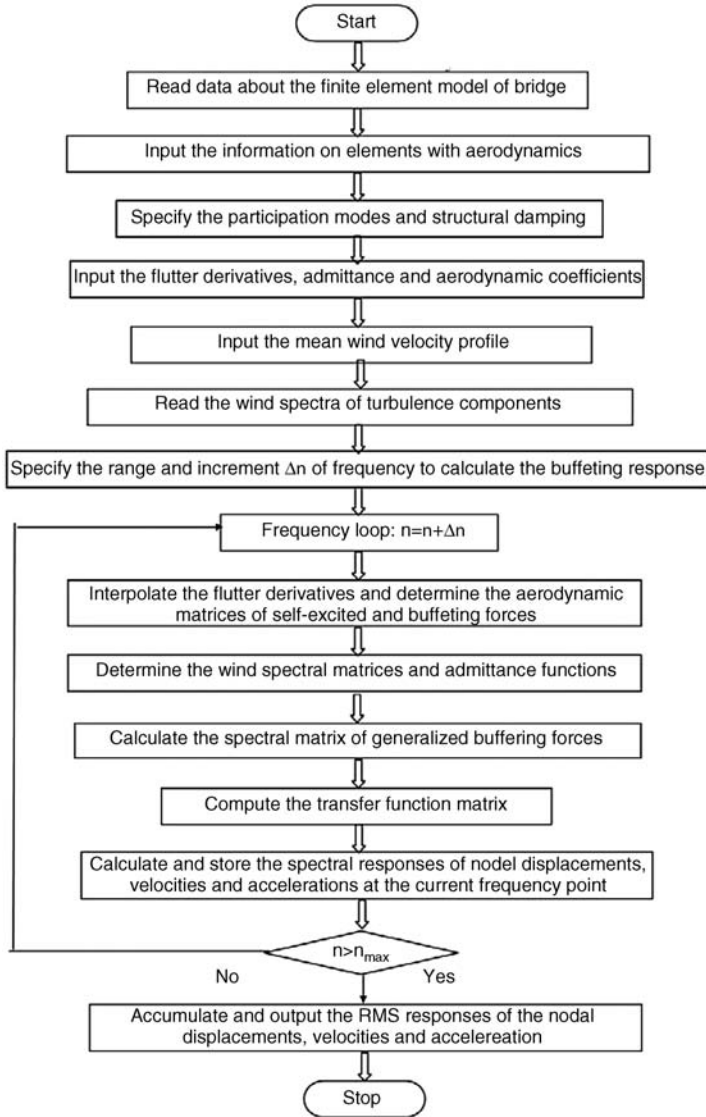


Figure 4.10 Flow chart of 3-D finite element based buffeting analysis for a long-span bridge.

ω_{up} is the upper cutoff frequency, with the condition that, when $\omega > \omega_{up}$, the value of $S^0(\omega)$ is trivial;
 ϕ_{ml} is the sequence of independent random phase angles, uniformly distributed over the interval $[0, 2\pi]$;

ω_{ml} is the double-indexing frequency:

$$\omega_{ml} = (l-1)\Delta\omega + \frac{m}{n}\Delta\omega \quad l = 1, 2, \dots, N \quad (4.86)$$

$H_{jm}(\omega)$ is a typical element of the matrix $H(\omega)$, which is defined as Cholesky decomposition of cross-spectral density matrix $S^0(\omega)$:

$$S^0(\omega) = H(\omega)H^{T*}(\omega) \quad (4.87)$$

$$H(\omega) = \begin{bmatrix} H_{11}(\omega) & 0 & \cdots & 0 \\ H_{21}(\omega) & H_{22}(\omega) & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ H_{n1}(\omega) & H_{n2}(\omega) & \cdots & H_{nn}(\omega) \end{bmatrix} \quad (4.88)$$

$\theta_{jm}(\omega)$ is the complex angle of $H_{jm}(\omega)$ and is given by:

$$\theta_{jm}(\omega) = \tan^{-1} \left\{ \frac{\text{Im}[H_{jm}(\omega)]}{\text{Re}[H_{jm}(\omega)]} \right\} \quad (4.89)$$

where $\text{Im}[H_{jm}(\omega)]$ and $\text{Re}[H_{jm}(\omega)]$ are the imaginary and real parts of the complex function $H_{jm}(\omega)$, respectively.

It has been proved that both ensemble and temporal auto- and cross- correlation functions of any sample function obtained by Equation 4.85 will approach to the target auto- and cross-correlation functions as $\Delta\omega \rightarrow 0$ and $N \rightarrow \infty$.

In order to avoid aliasing, the time step Δt has to obey the condition:

$$\Delta t \leq \frac{2\pi}{2\omega_{up}} \quad (4.90)$$

The period of the sample functions expressed by Equation 4.85 is:

$$T_0 = \frac{2\pi n}{\Delta\omega} = \frac{2\pi N}{\omega_{up}} \quad (4.91)$$

The ergodicity of the results of Equation 4.85 has been proved by Deodatis [26]. One can be confident that the one-dimensional stationary, multivariate, Gaussian stochastic process can be simulated quite well by means of Equation 4.85, when the cross-spectral density matrix is given and the values of the parameters N , ω_{up} and $\Delta\omega$ are properly chosen.

However, the number of wind velocity processes to be simulated is often very large in the buffeting analysis of long-span cable-supported bridges, when the stochastic wind loads on all major structural components such as bridge deck, cables, and towers shall be taken into consideration. It is difficult to do the simulation using the traditional method above, so an improved algorithm of the spectral representation method, which is considered suitable for the simulation of wind velocity filed on long-span cable-supported bridges, has been proposed by Ding *et al.* [27]. An interpolation approximation is introduced to simplify the computation of the lower triangular matrix with the Cholesky decomposition of the cross-spectral density matrix. Fast Fourier transform technique is used to further enhance the efficiency of computation in the simulation of stationary wind field.

4.7 Buffeting Analysis in the Time Domain

The buffeting response of a bridge induced by both self-excited and buffeting forces can also be calculated in the time domain. Compared to the frequency domain approach, time domain analysis offers the benefit of capturing the effects of non-linearities of both structural and aerodynamic origins and also

the influence of non-stationary features in the approaching wind in the analysis [17]. The theoretical background of time domain buffeting analysis is introduced in this section.

The self-excited forces per unit span can be expressed in terms of convolution integrals [28]:

$$L_{se}(t) = \frac{1}{2} \rho \bar{U}^2 \int_{-\infty}^t [I_{L_{seh}}(t-\tau)h(\tau) + I_{L_{sep}}(t-\tau)p(\tau) + I_{L_{seu}}(t-\tau)\alpha(\tau)] d\tau \quad (4.92a)$$

$$D_{se}(t) = \frac{1}{2} \rho \bar{U}^2 \int_{-\infty}^t [I_{D_{seh}}(t-\tau)h(\tau) + I_{D_{sep}}(t-\tau)p(\tau) + I_{D_{seu}}(t-\tau)\alpha(\tau)] d\tau \quad (4.92b)$$

$$M_{se}(t) = \frac{1}{2} \rho \bar{U}^2 \int_{-\infty}^t [I_{M_{seh}}(t-\tau)h(\tau) + I_{M_{sep}}(t-\tau)p(\tau) + I_{M_{seu}}(t-\tau)\alpha(\tau)] d\tau \quad (4.92c)$$

where $I_{()}$ is the impulse function of the self-excited forces, in which the subscript represents the corresponding force component.

The impulse functions can be obtained using the flutter derivatives measured from wind tunnel tests and the rational function approximation approach [17].

The relationship between the impulse functions and flutter derivatives in the frequency domain can be obtained by taking the Fourier transform of Equation 4.92 and comparing with Equation 4.33:

$$\bar{I}_{L_{seh}} = 2k^2(H_4^* + iH_1^*); \quad \bar{I}_{L_{sep}} = 2k^2(H_6^* + iH_5^*); \quad \bar{I}_{L_{seu}} = 2k^2b(H_3^* + iH_2^*) \quad (4.93a)$$

$$\bar{I}_{D_{seh}} = 2k^2(P_6^* + iP_5^*); \quad \bar{I}_{D_{sep}} = 2k^2(P_4^* + iP_1^*); \quad \bar{I}_{D_{seu}} = 2k^2b(P_3^* + iP_2^*) \quad (4.93b)$$

$$\bar{I}_{M_{seh}} = 2k^2b(A_4^* + iA_1^*); \quad \bar{I}_{M_{sep}} = 2k^2b(A_6^* + iA_5^*); \quad \bar{I}_{M_{seu}} = 2k^2b^2(A_3^* + iA_2^*) \quad (4.93c)$$

where the over-bar denotes the Fourier transform operator and $b = 0.5B$.

From the classical airfoil theory, the aerodynamic transfer functions in Equation 4.93 (the Fourier transform of the impulse functions) can be reasonably approximated by the rational functions. The frequency independent coefficients of the rational functions can then be determined by the non-linear least-squares method, using the flutter derivatives at different reduced frequencies. The impulse functions can be finally obtained by the inverse Laplace transform [17,29].

Because the aerodynamic derivatives are normally obtained from wind tunnel tests at discrete values of the reduced frequency k , some numerical methods are needed to extend these discrete values into continuous functions of the reduced frequency for time domain analysis.

Similarly, the buffeting forces per unit length can be expressed in terms of convolution integrals involving the aerodynamic impulse functions and fluctuating wind velocities:

$$L_b(t) = -\frac{1}{2} \rho \bar{U}^2 \int_{-\infty}^t \left[I_{L_{bu}}(t-\tau) \frac{u(\tau)}{\bar{U}} + I_{L_{bw}}(t-\tau) \frac{w(\tau)}{\bar{U}} \right] d\tau \quad (4.94a)$$

$$D_b(t) = -\frac{1}{2} \rho \bar{U}^2 \int_{-\infty}^t \left[I_{D_{bu}}(t-\tau) \frac{u(\tau)}{\bar{U}} + I_{D_{bw}}(t-\tau) \frac{w(\tau)}{\bar{U}} \right] d\tau \quad (4.94b)$$

$$M_b(t) = -\frac{1}{2} \rho \bar{U}^2 \int_{-\infty}^t \left[I_{M_{bu}}(t-\tau) \frac{u(\tau)}{\bar{U}} + I_{M_{bw}}(t-\tau) \frac{w(\tau)}{\bar{U}} \right] d\tau \quad (4.94c)$$

where $I_{()}$ is the impulse function of the buffeting force with the subscript representing the corresponding force component.

Similar to the self-excited forces, the impulse functions of the buffeting forces can be obtained by the inverse Laplace transform of the aerodynamic transfer functions, whereas the aerodynamic transfer functions of the buffeting forces can be related to aerodynamic coefficients and aerodynamic admittances as:

$$\bar{I}_{L_{bu}} = 4bC_L\chi_{L_{bu}}; \quad \bar{I}_{L_{bw}} = 2b(C'_L + C_D)\chi_{L_{bw}} \quad (4.95a)$$

$$\bar{I}_{D_{bu}} = 4bC_D\chi_{D_{bu}}; \quad \bar{I}_{D_{bw}} = 2bC'_D\chi_{D_{bw}} \quad (4.95b)$$

$$\bar{I}_{M_{bu}} = 8b^2C_M\chi_{M_{bu}}; \quad \bar{I}_{D_{bw}} = 4b^2C'_M\chi_{M_{bw}} \quad (4.95c)$$

Self-excited forces are usually assumed to be fully correlated. Thus, the self-excited force for an entire beam element can be expressed as:

$$L_{se}^e(t) = L_{se}^c(t)L \quad (4.96a)$$

$$D_{se}^e(t) = D_{se}^c(t)L \quad (4.96b)$$

$$M_{se}^e(t) = M_{se}^c(t)L \quad (4.96c)$$

where the superscript c indicates the center of the element.

The span-wise correlation of the buffeting forces in an element should be considered, and the buffeting forces for an entire beam element can be expressed as:

$$L_b^e(t) = L \int_0^t [J_{L_{bu}}(t-\tau)L_{bu}^c(\tau) + J_{L_{bw}}(t-\tau)L_{bw}^c(\tau)]d\tau \quad (4.97a)$$

$$D_b^e(t) = L \int_0^t [J_{D_{bu}}(t-\tau)D_{bu}^c(\tau) + J_{D_{bw}}(t-\tau)D_{bw}^c(\tau)]d\tau \quad (4.97b)$$

$$M_b^e(t) = L \int_0^t [J_{M_{bu}}(t-\tau)M_{bu}^c(\tau) + J_{M_{bw}}(t-\tau)M_{bw}^c(\tau)]d\tau \quad (4.97c)$$

where:

L_{bu}^c and L_{bw}^c are the first and second terms in Equation 4.94a at the center of the element;

J_0 is the impulse function, whose Fourier transform counterpart is referred to as the joint acceptance function:

$$\bar{J}_{L_{bu}}^2 = \frac{1}{L^2} \int_0^L \int_0^L coh_{L_{bu}}(x_1, x_2, \omega) dx_1 dx_2 \quad (4.98)$$

where:

$$coh_{L_{bu}}(x_1, x_2, \omega) = \frac{S_{L_{bu}}(x_1, x_2, \omega)}{S_{L_{bu}}^c(\omega)} \quad (4.99)$$

It is the span-wise coherence of the buffeting lift force component; $S_{L_{bu}}(x_1, x_2, \omega)$ is the cross-spectra between the buffeting forces at two different positions x_1 and x_2 ; and $S_{L_{bu}}^c(\omega)$ is the auto-spectral density of the buffeting force at the center of the element.

Once the local nodal self-excited forces and buffeting forces are obtained from Equation 4.96 and Equation 4.97, they can be converted into the global coordinate system using the coordinate transformation matrix. As a result, the global nodal buffeting force and self-excited force vectors $\mathbf{F}_{se}$ and $\mathbf{F}_b$ in Equation 4.62 can be obtained. The solution of the equation of motion (Equation 4.62) in the

time domain can be obtained by the Newmark-Beta method. Because the self-excited forces are dependent on the motion, iteration is needed for each time-step until certain convergence criterion is satisfied [29].

4.8 Effective Static Loading Distributions

4.8.1 Gust Response Factor and Peak Factor

Gust response factor G is a commonly used term in wind engineering and may be defined as the ratio of the expected maximum response of the structure in a defined period (e.g. ten minutes or one hour) to the mean or time-averaged response in the same time period [30–32]. Clearly, this term really has meaning only in stationary or near-stationary winds. The expected maximum response of the structure can be buffeting-induced maximum bending moment, deflection, or stress. It can be written as:

$$\hat{X} = \bar{X} + g\sigma_x \quad (4.100)$$

where:

$\hat{X}$ and $\bar{X}$ are the maximum and mean response of the structure;

σ_x is the standard deviation of the structural response;

g is a peak factor which depends on the frequency range of the response and the time interval for which the maximum value is calculated.

From Equation 4.100, the gust response factor can be expressed as:

$$G = \frac{\hat{X}}{\bar{X}} = 1 + g \frac{\sigma_x}{\bar{X}} \quad (4.101)$$

For a stationary Gaussian response to wind, Davenport [33] derived the following expression for the peak factor g :

$$g = \sqrt{2 \log_e(\nu T)} + \frac{0.577}{\sqrt{2 \log_e(\nu T)}} \quad (4.102)$$

where:

ν is the cycling rate or effective frequency for the response (often conservatively taken as the natural frequency of the structure);

T is the time period over which the maximum value is required.

For building structures, alongwind-induced responses are often estimated in terms of the sum of three components: mean component, background component and resonant component. Accordingly, the gust response factor can be expressed as:

$$G = \frac{\hat{X}}{\bar{X}} = 1 + 2g \frac{\sigma_u}{\bar{U}} \sqrt{B + R} \quad (4.103)$$

where:

σ_u is the standard deviation of turbulent wind in along-wind direction;

B is the background factor representing the quasi-static response caused by gusts below the first natural frequency of the structure;

R is the resonant factor describing the resonant response caused by gusts near the first natural frequency of the structure.

Equations 4.101 and 4.103 are used in many codes and standards for wind loading and, particularly, for alongwind dynamic loading of building structures. This is an approximate approach which works reasonably well for some structures and load effects, such as the base bending moment of tall buildings. However, in other cases it gives significant errors and should be used with caution [32].

4.8.2 Effective Static Loading Distributions

Effective static loading distributions are those loadings that produce the correct expected values of peak load effects generated by the fluctuating wind loading. For building structures, effective static peak loading distributions can be derived from three components: mean components, background component and resonant component. The main advantage of the effective static loading distribution approach is that the distributions can be applied to a static structural analysis computer program for use in final structural design and can be used relatively easily by engineers and incorporated in design codes and standards. The following introduction on this topic refers to Holmes's book [32].

By taking alongwind (drag) response of a long-span bridge deck as an example, and regarding the bridge deck as a continuous beam, the mean wind force per unit length on the bridge deck can be written as:

$$\bar{f}(x) = \frac{1}{2} \rho \bar{U}^2(x) B(x) C_D(x) \quad (4.104)$$

The mean value of any load effect can be obtained by integrating the local force with the influence line along the length of the bridge deck. If the purpose is to derive an effective static loading, Equation 4.104 is already in the correct form.

The background wind loading is the quasi-static loading produced by fluctuations due to turbulence, but with frequencies too low to excite any resonant response. The load-response correlation formula derived by Kasperski and Niemann [34] can be used to derive the effective background fluctuating loading distribution. In the form of a continuous distribution, this term can be written as:

$$f_B(x) = g_B \rho(x) \sigma_p(x) \quad (4.105)$$

where:

g_B is the peak factor for the background response;

$\sigma_p(x)$ is the standard deviation of the fluctuating load at position x ;

$\rho(x)$ is the correlation coefficient between the fluctuating load at position x and the load effect of interest [32]:

$$\rho(x) = \frac{\int_s^L \overline{f'(x) f'(x_1)} I_r(x_1) B(x_1) dx_1}{\sqrt{\overline{f'^2(x)}} \cdot \sqrt{\int_s^L \int_s^L \overline{f'(x_1) f'(x_2)} I_r(x_1) I_r(x_2) B(x_1) B(x_2) dx_1 dx_2}} \quad (4.106)$$

where:

$I_r(x)$ is the influence line for the load effect r , as a function of position x ;

$f'(x)$ is the fluctuating alongwind force on the bridge deck;

$\overline{f'(x) f'(x_1)}$ is the covariance for the fluctuating alongwind forces at positions x and x_1 .

The equivalent load distribution for the resonant response in the first mode of vibration can be represented as a distribution of inertial forces over the length of the bridge deck:

$$f_R(x) = g_R m(x) (2\pi n_1)^2 \sqrt{a'^2} \phi_1(x) \quad (4.107)$$

where:

$f_R(x)$ is the equivalent load distribution for the resonant response at position x ;

g_R is the peak factor for resonant response;

$m(x)$ is the mass per unit length at position x ;

n_1 is the first mode natural frequency;

$\sqrt{a'^2}$ is the standard deviation of modal coordinate (resonant contribution only);

$\phi_1(x)$ is the first mode shape.

Determining the standard deviation of modal coordinate due to the resonant contribution only requires knowledge of the spectral density of the fluctuating forces, the correlation of those forces at the natural frequency and the modal damping and stiffness. The total effective loading distribution can be obtained by:

$$f_c(x) = \bar{f}(x) + w_{back} f_B(x) + w_{res} f_R(x) \quad (4.108)$$

where w_{back} and w_{res} are the weight factors for background and resonant response, respectively.

Assuming that the fluctuating background and resonant components are uncorrelated with each other, the weight factors can be calculated by:

$$|w_{back}| = \frac{g_B \sigma_{r,B}}{\sqrt{g_B^2 \sigma_{r,B}^2 + g_R^2 \sigma_{r,R}^2}} \quad (4.109a)$$

$$|w_{res}| = \frac{g_R \sigma_{r,R}}{\sqrt{g_B^2 \sigma_{r,B}^2 + g_R^2 \sigma_{r,R}^2}} \quad (4.109b)$$

where:

$\sigma_{r,B}$ and $\sigma_{r,R}$ are the standard deviation of the background response and the resonant response, respectively;

the subscript r means the load effect consistent with the response.

The method of effective static loading distribution approach described above can be applied to common bridges in principle. However, for long-span cable-supported bridges, it is often the case that several modes of vibration must be considered. Holmes [35] proposed a method of combining inertial force distributions from more than one resonant mode of vibration. By neglecting the background response, the total effective loading distribution for the peak load effect, r , is [32]:

$$f_{eff}(x) = \bar{f}(x) + m(x) \sum_{j=1}^N W_j \phi_j(x) \quad (4.110)$$

where W_j is the peak inertial force in the mode j .

4.9 Case Study: Stonecutters Bridge

4.9.1 Dynamic and Aerodynamic Characteristics of Stonecutters Bridge

The finite element modeling technique for long-span cable-supported bridges and its application for developing a 3D finite element model of the Stonecutters Bridge have been introduced in Chapter 3. The frequency domain flutter and buffeting analyses of the Stonecutters Bridge are introduced in this section as a case study using that model [16,36].

The first 50 natural modes of the bridge were obtained from the modal analysis of the finite element model. The first 50 natural frequencies, as well as the corresponding mode shape descriptions, are listed in Table 4.2. It can be seen from this table that the first 50 natural frequencies of the bridge range from 0.145 Hz to 1.447 Hz. The lowest natural frequency of 0.145 Hz corresponds to the first horizontal mode of vibration, in which the motion of the bridge deck is almost in symmetry in the main span. The first vertical vibration mode dominated by the bridge deck is almost symmetric in the main span at a natural frequency of 0.201. The first torsional mode of vibration dominated by the bridge deck is almost symmetric in the main span at a natural frequency of 0.425 Hz.

The aerodynamic coefficients of the Stonecutters bridge deck without traffic were obtained from the section model tests in a wind tunnel. The drag, lift, and moment coefficients for the cross section of the entire bridge deck at zero degree of attack angle are $C_D = 0.073$, $C_L = -0.155$ and $C_M = -0.018$. The derivatives of the drag, lift, and moment aerodynamic coefficients with wind angle of attack at zero degree are $dC_D/d\alpha = 0.069$, $dC_L/d\alpha = 2.510$ and $dC_M/d\alpha = 0.539$. These aerodynamic coefficients are normalized by the overall deck width of 53.3 m.

The drag coefficient of the bridge tower is taken as $C_D = 0.9$ along the entire height of the tower and is normalized by the actual width of the tower perpendicular to the wind direction. The drag coefficient of the piers is taken as $C_D = 1.1$ for wind perpendicular to the bridge longitudinal axis and is normalized by the actual width of the piers perpendicular to the wind direction. The drag coefficient of the stay cables is taken as $C_D = 0.8$ for wind perpendicular to the bridge longitudinal axis and is normalized by the actual diameter of the stay cables. As the cross sections of the stay cables, towers and piers are symmetric, their aerodynamic lift and moment coefficients for wind perpendicular to the bridge longitudinal axis are taken as zero.

The flutter derivatives of the Stonecutters Bridge deck without traffic were also obtained from the section model tests in a wind tunnel. Only the flutter derivatives H_i^* and A_i^* ($i = 1 - 4$) are available, and they are listed in Table 4.3 and plotted in Figure 4.11. Since the flutter derivatives related to the lateral motion of the bridge deck and the flutter derivatives H_5^* , H_6^* , A_5^* , A_6^* are not available, they are calculated based on the quasi-steady theory in terms of the aerodynamic coefficients:

$$P_1^* = -\frac{1}{K}C_D, \quad P_2^* = \frac{1}{2K}C_D', \quad P_3^* = \frac{1}{2K^2}C_D' \quad (4.111a)$$

$$P_5^* = \frac{1}{2K}C_D', \quad H_5^* = \frac{1}{K}C_L, \quad A_5^* = -\frac{1}{K}C_M \quad (4.111b)$$

$$P_4^* = P_6^* = H_6^* = A_6^* = 0 \quad (4.111c)$$

The flutter derivatives and the aerodynamic coefficients mentioned above are assumed to be uniform along the bridge deck in the both flutter and buffeting analysis.

4.9.2 Flutter Analysis of Stonecutters Bridge

The aerodynamically coupled flutter analysis is carried out in the complex frequency domain using in-house software packages with theoretical background, as introduced in Section 4.4. By using the modal coordinates of the bridge, the governing equation of the bridge for the flutter analysis is converted into a

Table 4.2 Dynamic characteristics of Stonecutters Bridge

Mode no.	Natural frequency (Hz)	Period (sec)	Mode shape description
1	0.145	6.875	1.symmetric horizontal, deck
2	0.190	5.260	1.asymmetric horizontal, towers
3	0.197	5.073	1.symmetric horizontal, towers
4	0.201	4.968	1.symmetric vertical, deck
5	0.246	4.072	1.asymmetric vertical, deck
6	0.302	3.311	2.asymmetric vertical, deck
7	0.311	3.217	2.symmetric vertical, deck
8	0.361	2.772	1.asymmetric horizontal, deck
9	0.376	2.660	3.asymmetric vertical, deck
10	0.422	2.372	3. symmetric vertical, deck
11	0.425	2.354	1.symmetric torsional, deck
12	0.444	2.250	Horizontal, deck and piers, west side span
13	0.452	2.214	Horizontal, deck and piers, east side span
14	0.490	2.040	4.asymmetric vertical, deck
15	0.525	1.903	2.symmetric horizontal, towers
16	0.535	1.869	4.symmetric vertical, deck
17	0.579	1.728	2.asymmetric horizontal, towers
18	0.591	1.693	1.asymmetric torsional, deck
19	0.592	1.688	5.symmetric vertical, deck
20	0.620	1.613	5.asymmetric vertical, deck
21	0.682	1.466	2.symmetric torsional, deck
22	0.694	1.440	6.symmetric vertical, deck
23	0.729	1.372	Symmetric horizontal and torsional, deck
24	0.768	1.303	6.asymmetric vertical, deck
25	0.827	1.209	2.asymmetric torsional, deck
26	0.842	1.188	7.symmetric vertical, deck
27	0.900	1.111	Horizontal, deck and piers, west & east side spans
28	0.907	1.103	Horizontal, deck and piers, west & east side spans
29	0.912	1.096	7.asymmetric vertical, deck
30	0.948	1.055	3.symmetric torsional, deck
31	0.969	1.032	Asymmetric horizontal, deck and tower
32	0.973	1.027	8.symmetric vertical, deck
33	1.003	0.998	1.asymmetric longitudinal, tower
34	1.034	0.967	Symmetric longitudinal, deck and tower
35	1.049	0.953	8.asymmetric vertical, deck
36	1.072	0.933	3.asymmetric torsional, deck
37	1.129	0.886	9.symmetric vertical, deck
38	1.132	0.884	Symmetric horizontal, deck
39	1.193	0.839	4.symmetric torsional, deck
40	1.216	0.822	9.asymmetric vertical, deck
41	1.245	0.803	Asymmetric horizontal, deck and tower
42	1.254	0.797	10. symmetric vertical, deck
43	1.265	0.791	10.asymmetric vertical, deck
44	1.298	0.771	4.asymmetric torsional, deck
45	1.314	0.761	Symmetric vertical, towers and deck
46	1.345	0.744	Asymmetric vertical, towers and deck
47	1.364	0.733	11.symmetric vertical, deck
48	1.379	0.725	5.symmetric torsional, deck
49	1.407	0.711	11.asymmetric vertical, deck
50	1.447	0.691	Symmetric horizontal, deck and tower

Table 4.3 Flutter derivatives of the Stonecutters Bridge deck

(a) Flutter derivatives $H_1^*, H_4^* A_1^*, A_4^*$				
U/fB	H_1^*	H_4^*	A_1^*	A_4^*
0.00	0.000	0.000	0.000	0.000
2.35	-0.280	-0.020	0.040	0.044
3.50	-0.390	0.000	0.110	0.080
4.71	-0.550	-0.080	0.170	0.100
5.88	-0.760	-0.170	0.240	0.120
6.93	-1.020	-0.330	0.310	0.140
8.08	-1.310	-0.430	0.360	0.160
9.18	-1.570	-0.470	0.400	0.124
10.44	-1.830	-0.450	0.430	0.064
11.56	-2.040	-0.470	0.460	0.050
13.00	-2.370	-0.500	0.500	0.036

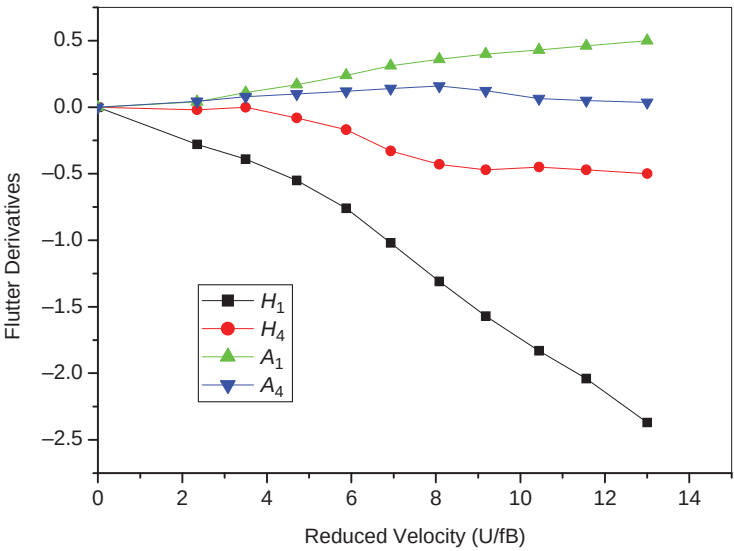
(b) Flutter derivatives $H_2^*, H_3^* A_2^*, A_3^*$				
U/fB	H_2^*	H_3^*	A_2^*	A_3^*
0.00	0.000	0.000	0.000	0.000
2.02	-0.210	-0.320	-0.037	-0.014
2.88	-0.390	-0.430	-0.083	0.024
3.81	-0.440	-0.590	-0.119	0.069
5.00	-0.360	-0.900	-0.172	0.131
6.32	-0.180	-1.440	-0.237	0.204
7.41	0.120	-2.110	-0.339	0.308
9.52	0.380	-2.850	-0.403	0.415
10.93	0.580	-3.920	-0.419	0.569
13.44	-2.750	-6.600	-0.543	0.956

complex characteristic equation with only two variables. A single parameter-searching method is then used to find the lowest critical wind speed without choosing participating modes beforehand. The major participating modes of vibration causing the flutter instability and the phase angles between the participating modes of vibration can also be found.

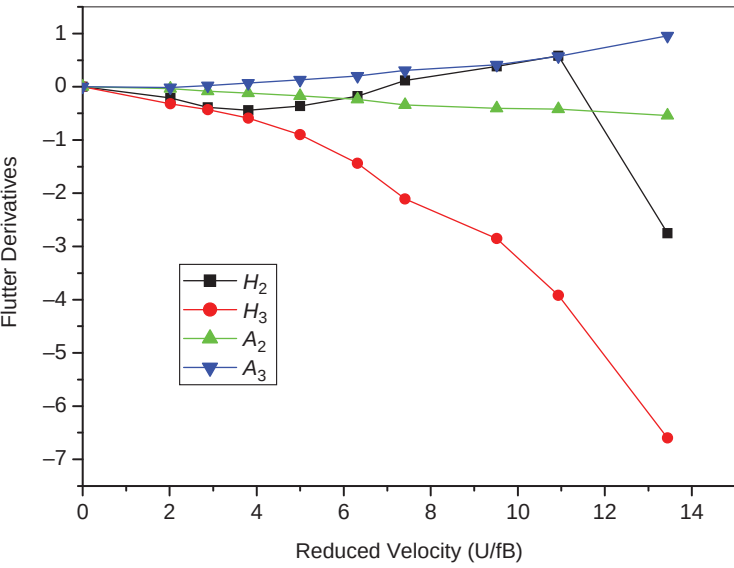
The effects of the towers, pier shafts and stay cables on the flutter instability of the bridge are also taken into consideration. The flutter derivatives for these bridge components are calculated based on the quasi-steady theory. Since the cross-sections of the tower, pier shafts and stay cables are basically circular, only the drag coefficients are needed to calculate the flutter derivatives required. Since the method used for flutter analysis has no limit on the number of modes, the first 50 modes of vibration of the bridge are employed as the participating modes for the flutter analysis of the bridge.

The lowest critical wind speed for the bridge is computed as being 230.4 m/s at a reduced velocity of 13.08 and a flutter frequency of 0.331 Hz. Such a high wind speed will not appear in practice. Therefore, the computed critical wind speed is of theoretical interest only, and it indicates no flutter instability problem for the concerned bridge.

Corresponding to the lowest critical wind speed or the flutter frequency, a flutter eigenvector can be found. From this, one may observe the distribution of modal motion and modal energy over all the participating modes of vibration and the distribution of modal phase angle among all the participating modes of vibration. From these distributions, the dominant modes of vibration causing the flutter instability can then be identified.



(a) Flutter derivatives H_1^* , H_4^* , A_1^* , A_4^*



(b) Flutter derivatives H_2^* , H_3^* , A_2^* , A_3^*

Figure 4.11 Flutter derivatives of Stonecutters bridge deck without traffic.

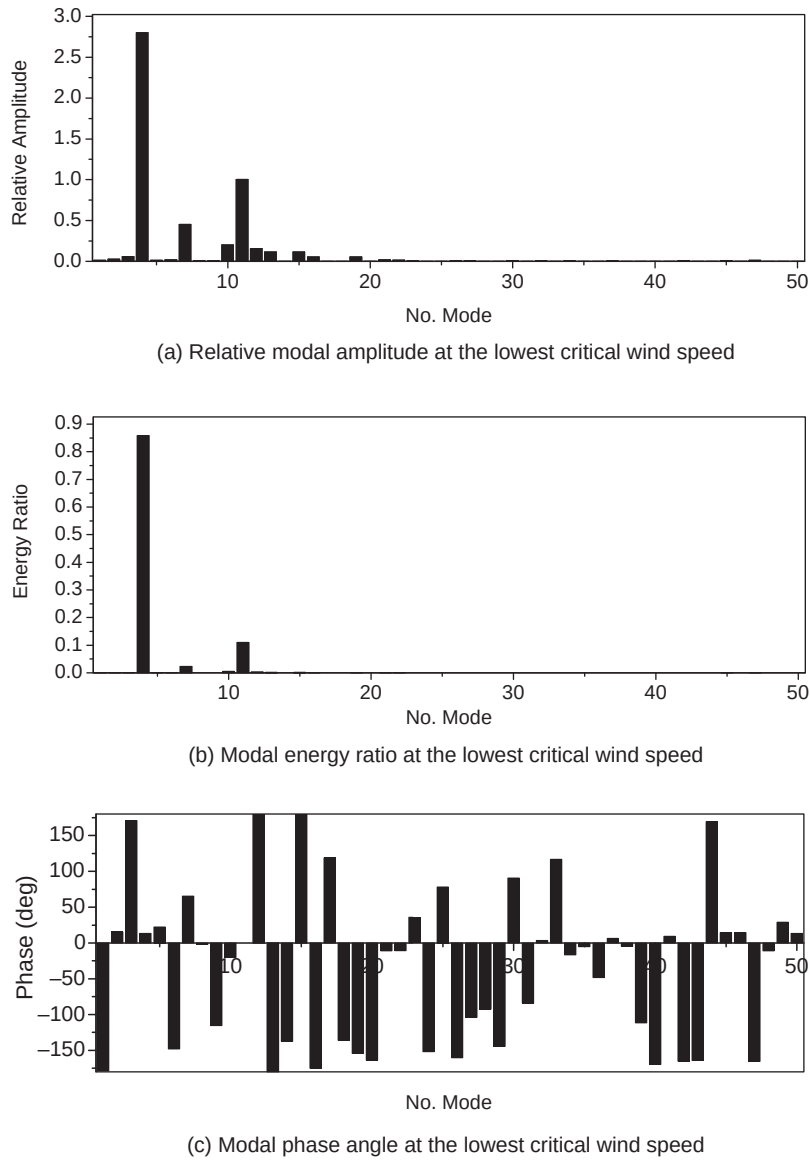


Figure 4.12 Modal information of the bridge at the lowest critical wind speed.

Figure 4.12 shows the distribution of relative modal amplitude, modal energy ratio and modal phase angle over the 50 participating modes of vibration, respectively. It is noted that vibration modes 4 (first symmetric vertical mode) and 11 (first symmetric torsional mode) are the two major participating modes dominating the flutter instability of the bridge, with almost the same phase angle. Although the lateral vibration modes of the bridge deck also participate in the flutter motion, the degree of their participations is very small.

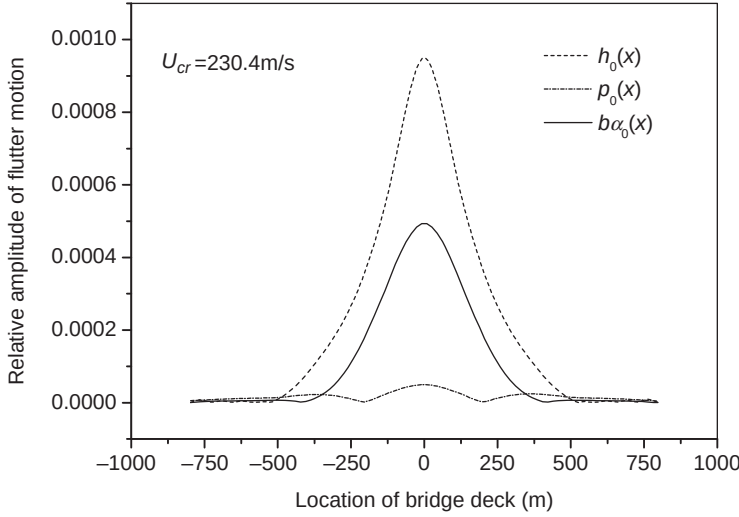


Figure 4.13 Relative amplitudes of deck motion along the bridge longitudinal axis.

Apart from the modal information, Figure 4.13 shows the relative amplitudes of the vertical, lateral and torsional displacement responses (flutter motion) of the bridge deck along the bridge longitudinal axis at the lowest critical wind speed. Note again that the vertical and torsional vibrations dominate the flutter motion of the bridge deck. More details can be found in the literature [16].

4.9.3 Buffeting Analysis of Stonecutters Bridge

The 3-D finite element-based buffeting analysis is to determine the total buffeting response of the Stonecutters Bridge to wind loads specified in the design memorandum [36]. The mean wind velocity is considered to be perpendicular to the bridge longitudinal axis, with zero angle of attack from either the southwest (SW) direction or the northeast (NE) direction. For the sake of brevity, only the NE direction, in which the terrain is considered as a overland fetch, is concerned in this case study.

The 3-D finite element-based coupled buffeting response analysis of the Stonecutters Bridge is carried out in the frequency domain, with the theoretical background introduced in Section 4.5. Both structural and aeroelastic couplings between modes of vibration and the interaction between the bridge deck, towers and cables are naturally included in the buffeting analysis. Varying structural properties along the bridge deck are taken into consideration. The structural damping ratio of the bridge is assumed to be 0.36% for each mode of vibration in the buffeting analysis, according to the design specification. This level of structural damping is considered a realistic value. The first 50 modes of vibration (excluding local cable modes) are included in the buffeting analysis.

The one-hour mean wind velocity profile $\bar{U}(z)$ in m/s for wind from the NE direction is taken as:

$$\bar{U}(z) = 0.66 \cdot 35 \cdot \left(\frac{z}{10}\right)^{0.29} \quad (4.112)$$

The ten-minute mean wind velocity profile $\bar{U}(z)$ in m/s for wind from the NE direction is taken as:

$$\bar{U}_{10}(z) = \bar{U}_{10}(10) \cdot \left(\frac{z}{10}\right)^{0.29} \quad (4.113)$$

The ten-minute mean wind velocity $\bar{U}_{10}(10)$ at 10 m height above ground is defined as the reference wind velocity, and its basic value is 24.5 m/s for wind from the NE direction. The longitudinal turbulent intensity I_u for wind from the NE direction is given by the following expression:

$$I_u = \frac{\sigma_u}{\bar{U}_{10}(z)} = 0.37 \cdot \left(\frac{10}{z}\right)^{0.29} \quad (4.114)$$

where σ_u is also assumed to be constant with height.

For lateral (v) and vertical (w) turbulence intensities, the following relationships are assumed, based on analysis of typhoon data recorded at the Tsing Ma Bridge for the overland fetch:

$$I_v = 1.00 \cdot I_u; \quad I_w = 0.75 \cdot I_u \quad (4.115)$$

The power spectra of the longitudinal (u), lateral (v) and vertical (w) turbulent wind components are defined according to the von Kármán spectral model.

For longitudinal turbulent wind component:

$$\frac{f \cdot S_{uu}(f)}{\sigma_u^2} = \frac{4 \frac{L_u \cdot f}{\bar{U}_{10}(z)}}{\left[1 + 70.8 \cdot \left(\frac{L_u \cdot f}{\bar{U}_{10}(z)}\right)^2\right]^{5/6}} \quad (4.116)$$

For lateral and vertical turbulent wind components:

$$\frac{f \cdot S_{vv,ww}(f)}{\sigma_{v,w}^2} = \frac{4 \frac{L_{v,w} \cdot f}{\bar{U}_{10}(z)} \left[1 + 755 \cdot \left(\frac{L_{v,w} \cdot f}{\bar{U}_{10}(z)}\right)^2\right]}{\left[1 + 283 \cdot \left(\frac{L_{v,w} \cdot f}{\bar{U}_{10}(z)}\right)^2\right]^{11/6}} \quad (4.117)$$

where:

$S_{ii}(f)$ is the wind energy spectrum for turbulent wind components i ;

f is the frequency in Hz;

L_i is the integral length scale for turbulent wind component i .

Turbulence integral scale is specified in the design memorandum as:

$$L_i = L_{i50} \cdot \left(\frac{z}{50}\right)^{0.45} \quad (4.118)$$

where L_{i50} represents the length scale at 50 m height above ground for turbulent wind component i , and $L_{u50} = 160$ m, $L_{v50} = L_{u50}/3$, $L_{w50} = L_{u50}/6$ are specified for all wind directions.

The root coherence, which defines the statistical dependency between the turbulent wind components at two different points, is given by the following expression:

$$\sqrt{coh} = \exp\left(-C \cdot \frac{D \cdot f}{\bar{U}_{10}(z)}\right) \quad (4.119)$$

Table 4.4 Values of the decay factor C

	Lateral separation	Longitudinal separation	Vertical separation
Longitudinal turbulent wind component	8	2	8
Lateral turbulent wind component	4	4	8
Vertical turbulent wind component	8	4	4

where the parameter C is the decay factor; and the parameter D is the distance between the two points. The values of the decay factor C for all 9 coherences are listed in Table 4.4.

Since the basic value of the ten-minute mean wind speed at 10 m height above ground is 24.4 m/s for wind from the N-E direction, the corresponding basic value of the ten-minute mean wind speed at the middle point of the main span of the bridge deck is 46 m/s. The mean wind response of the complete bridge is then computed for the deck-reference mean wind velocity of 46 m/s.

The mean wind displacement responses of the bridge are computed, and those at the key locations are listed in Table 4.5. The positions of the key locations are the same as shown in Figure 3.14. It is noted that the mean lateral, vertical and torsional displacement responses of the windward deck at the middle point of main span are -0.449 m, -0.202 m and -0.002 rad, respectively. The mean lateral, vertical and torsional displacements of the leeward deck at the middle point of main span are -0.449 m, -0.268 m and -0.002 rad, respectively. The mean lateral and vertical displacement responses for wind from the NE direction are relatively smaller compared with those for wind from the SW direction (as introduced in Chapter 3), because of smaller design mean wind speed. The mean lateral displacement at the top of the tower reaches -0.869 m. For other mean wind speeds, the mean wind displacement responses of the bridge at the key locations are also computed. Figure 4.14 shows the mean displacement responses of the windward deck at the middle of main span versus NE mean wind speed.

The coupled buffeting responses of the bridge are computed using the mean wind speed at the middle point of the main span of the bridge deck as a reference mean wind speed of 46 m/s as well. By using the Sears function as admittance functions for the bridge deck, the buffeting peak responses of the bridge are computed and the peak displacement and acceleration responses at the key locations are listed in Table 4.6 and Table 4.7, respectively, for the basic mean wind velocity of 46 m/s.

The peak buffeting response refers to the standard deviation buffeting response multiplied by a peak factor. The peak factor of the buffeting response is taken as 3.5 for the purposes of this case study. The total wind response, then, refers to the mean wind response plus the peak buffeting response.

Table 4.5 Mean wind displacement responses of the bridge at key locations (NE direction, $V_{dref} = 46$ m/s)

Location	Longitudinal (m)	Vertical (m)	Lateral (m)	Around x -axis (rad)	Around y -axis (rad)	Around z -axis (rad)
D1-windward	$-1E-04$	-0.202	-0.449	-0.002	$-2E-06$	$-1E-06$
D1-leeward	$-4E-05$	-0.268	-0.449	-0.002	$-2E-06$	$-7E-05$
D2-windward	0.014	-0.054	-0.294	-0.003	$1E-03$	$-6E-04$
D2-leeward	-0.013	-0.138	-0.294	-0.002	$1E-03$	$-6E-04$
D5-windward	0.006	0.013	-0.033	$-9E-04$	$2E-04$	$1E-04$
D5-leeward	$2E-04$	-0.012	-0.033	$-9E-04$	$2E-04$	$-7E-05$
T1-tower-top	0.046	-0.004	-0.869	-0.006	$5E-05$	$-2E-04$

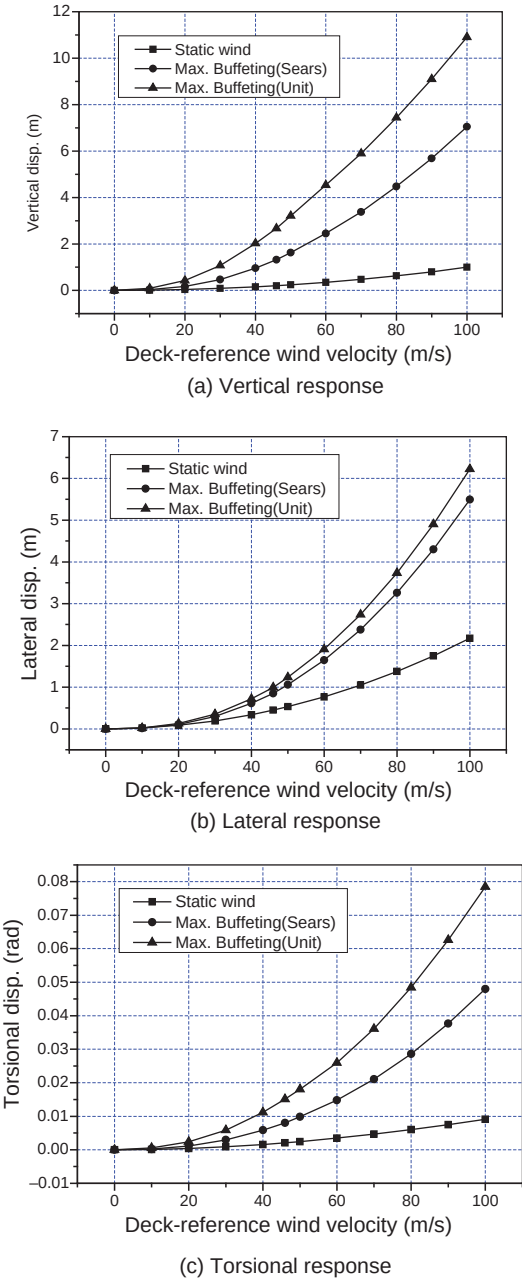


Figure 4.14 Displacement responses of the windward deck at the middle of main span.

Table 4.6 Buffeting peak displacement responses of the bridge at key locations (NE direction, $V_{dref} = 46$ m/s, Sears admittance)

Location	Longitudinal (m)	Vertical (m)	Lateral (m)	Around x -axis (rad)	Around y -axis (rad)	Around z -axis (rad)
D1-windward	0.051	1.322	0.848	0.008	7E-04	0.008
D1-leeward	0.051	1.219	0.848	0.008	7E-04	0.008
D2-windward	0.057	0.505	0.529	0.006	0.002	0.004
D2-leeward	0.057	0.444	0.529	0.007	0.002	0.004
D5-windward	0.053	0.029	0.088	0.002	3E-04	2E-04
D5-leeward	0.053	0.029	0.088	0.002	3E-04	2E-04
T1-tower-top	0.257	0.009	1.519	0.009	1E-04	0.002

Note that the lateral, vertical and torsional buffeting peak displacement responses of the windward deck at the middle point of main span are 0.848 m, 1.322 m and 0.008 rad, respectively. The lateral, vertical and torsional buffeting peak displacement responses of the leeward deck at the middle point of main span are 0.848 m, 1.219 m and 0.008 rad, respectively. The lateral buffeting peak displacement response at the top of the tower reaches 1.519 m. The lateral and vertical buffeting peak acceleration responses of the windward deck at the middle point of main span are 0.783 m/s^2 and 2.111 m/s^2 , respectively, while the lateral buffeting peak acceleration response at the top of the tower reaches 2.146 m/s^2 .

The buffeting peak displacement and acceleration responses of the bridge at the key locations are also computed for other wind speeds from the NE direction using the Sears function and the unit admittance, respectively. Figures 4.14 and 4.15 display the variations of the buffeting peak (maximum) displacement and acceleration responses, respectively, of the windward deck at the middle of main span, versus NE mean wind speed.

The total wind response of the complete bridge is equal to the sum of the absolute mean wind response and the buffeting peak response of the bridge. For the Sears function used as admittance functions for the bridge deck under the N-E wind, the total wind responses of the bridge are computed. The total displacement responses at the key locations are listed in Table 4.8 for the basic mean wind velocity of 46 m/s.

Table 4.7 Buffeting peak acceleration responses of the bridge at key locations (NE direction, $V_{dref} = 46$ m/s, Sears admittance)

Location	Longitudinal (m/s^2)	Vertical (m/s^2)	Lateral (m/s^2)	Around x -axis (rad)	Around y -axis (rad)	Around z -axis (rad)
D1-windward	0.186	2.111	0.783	0.034	0.002	0.028
D1-leeward	0.186	2.130	0.783	0.034	0.002	0.028
D2-windward	0.195	1.582	0.546	0.026	0.003	0.025
D2-leeward	0.195	1.609	0.546	0.026	0.003	0.025
D5-windward	0.221	0.048	0.180	0.003	0.001	0.002
D5-leeward	0.221	0.049	0.180	0.003	0.001	0.002
T1-tower-top	0.493	0.020	2.146	0.016	5E-04	0.008

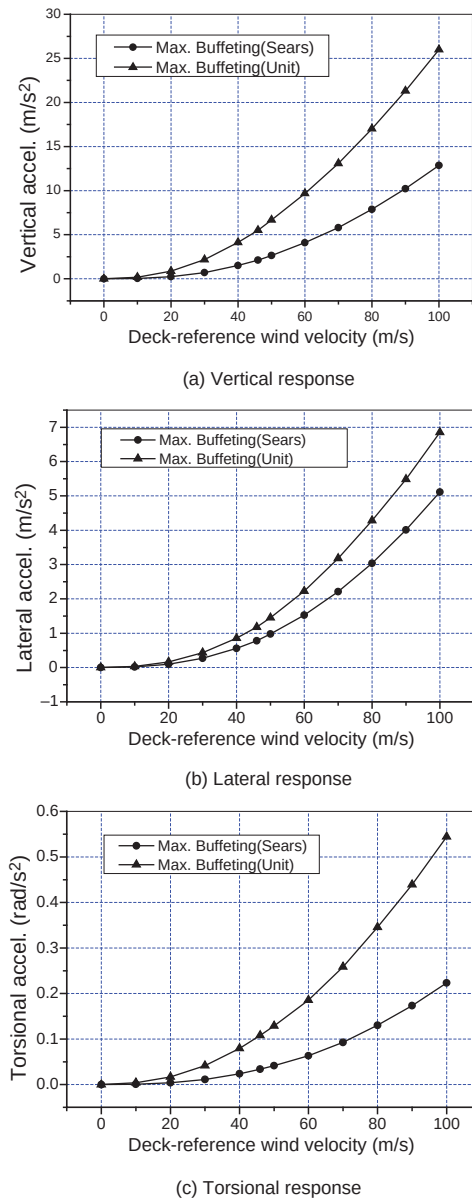


Figure 4.15 Acceleration responses of the windward deck at the middle of main span versus N-E mean wind speed.

Table 4.8 Total displacement responses of the bridge at key locations (NE direction, $d_{ref} = 46$ m/s, Sears admittance)

Location	Longitudinal (m)	Vertical (m)	Lateral (m)	Around x-axis (rad)	Around y-axis (rad)	Around z-axis (rad)
D1-windward	0.051	1.523	1.297	0.010	7E-04	0.008
D1-leeward	0.051	1.487	1.297	0.010	7E-04	0.008
D2-windward	0.071	0.559	0.822	0.009	0.003	0.005
D2-leeward	0.070	0.583	0.822	0.009	0.003	0.005
D5-windward	0.059	0.042	0.122	0.003	5E-04	4E-04
D5-leeward	0.053	0.041	0.122	0.003	5E-04	3E-04
T1-tower-top	0.303	0.013	2.388	0.015	2E-04	0.003

4.10 Notations

a_i	Parameters to be determined by experiments
$\sqrt{a'^2}$	Standard deviation of modal coordinate
A	State matrix
A_{bu}	Global buffeting force matrices corresponding to the longitudinal wind fluctuation
A_{bw}	Global buffeting force matrices corresponding to the vertical wind fluctuation
A_{bu}^e	Buffeting force matrices of element corresponding to the longitudinal wind fluctuation
A_{bw}^e	Buffeting force matrices of element corresponding to the vertical wind fluctuation
A_i^* ($i = 1 - 6$)	Aerodynamic derivatives
A_{se}	Complex aeroelastic matrix of the bridge
b	Half of the plate width
B	(i) Typical chord or deck width dimension (ii) Matrix of interpolated function (Equation 4.66) (iii) Background factor representing the quasi-static response caused by gusts below the first natural frequency of the structure (Equation 4.103)
c	Damping coefficient of the structure
$\sqrt{coh}$	Root coherence
C	Decay factor
C	Structural damping matrix
C_D	Drag force coefficient
C_L	Lift force coefficient
C_M	Moment coefficient
$C_L(K)$	Function of the reduced frequency K at lock-in
$C(k)$	Theodorsen cyclical function
C_{rs}	Complex flutter derivatives of self-excited forces
C^e	Damping matrix of the coupled wind-structure system
D	(i) Cylinder diameter (Equation 4.1) (ii) Characteristic dimension (Equation 4.2) (iii) Bridge deck width (Equation 4.10) (iv) Distance between the two points (Equation 4.119)
$D(\alpha)$	Aerodynamic drag force
E	Total energy in the characteristic motion of the bridge at the lowest critical wind speed
f	Frequency
$f_B(x)$	Load-response correlation formula

$f_c(x)$	Total effective loading distribution
f_{st}	Vortex shedding frequency
f_n, f_s	Vortex-shedding frequency or the natural frequency
$f_R(x)$	Equivalent load distribution for the resonant response at position x
$\bar{f}(x)$	Mean wind force per unit length on the bridge deck
$f'(x)$	Fluctuating along-wind force on the bridge deck
$\overline{f'(x)f'(x_1)}$	Covariance for the fluctuating along-wind forces at positions x and x_1
$\mathbf{F}$	Nodal equivalent force vector
$\mathbf{F}_{se}$	Global nodal self-excited force vector
$\mathbf{F}_b$	Global nodal buffeting force vector
F_s	Static force
$F(K)$	Real part of Theodorsen cyclical function
$F_y(\alpha)$	Vertical force in the y -direction
g	Peak factor
g_B	Peak factor for background response
g_R	Peak factor for resonant response
G	Gust response factor
$G(K)$	Imaginary part of Theodorsen cyclical function
h	Vertical displacement of structure or bridge deck
$H_i^*(i = 1 - 6)$	Aerodynamic derivatives
$H(\omega)$	(i) Transfer function matrix (Equation 4.77) (ii) Cholesky decomposition of cross-spectral density matrix $S^0(\omega)$ (Equation 4.86)
I	(i) Mass moment of inertia of the structure (Equation 4.22) (ii) Unit matrix (Equation 4.31)
I_0	Impulse function of the self-excited forces
$I_r(x)$	Influence line for the load effect r
I_u, I_v, I_w	Longitudinal, lateral and vertical turbulence intensity, respectively
J_0	Impulse function whose Fourier transform counterparts are referred to as the joint acceptance function
k	Stiffness coefficient
k_α	Total stiffness for the wind-structure system
$\mathbf{K}$	Structural stiffness matrix
K	Reduced frequency
K^e	Stiffness matrix of the coupled wind-structure system
$L(\alpha)$	Aerodynamic lift force
L_{se}	Self-excited lift force
$\mathbf{L}_{vs}$	Vortex-induced lift force
L_{i50}	Integral length scale at 50m height above ground for turbulent wind component i
L_u, L_v, L_w	Longitudinal, lateral and vertical turbulence integral scale, respectively
$\mathbf{M}$	Structural mass matrix
M	Aerodynamic pitching moment
M_{se}	Self-excited moment
m	(i) Mass of the structure (ii) Number of participating mode
N	Sufficiently large number
N_p	Number of points in the wind field
n_1	First mode natural frequency
p	Lateral displacement of the bridge deck
$P_i^*(i = 1 - 6)$	Aerodynamic derivatives
q	Generalized modal response

$\mathbf{Q}_b$	Generalized buffeting force vector
Re	Reynolds number
R	(i) Complex modal response amplitude vector (Equation 4.41) (ii) Resonant factor describing the resonant response (Equation 4.103)
s	Complex frequency
S	(i) PSD matrix (Equation 4.78) (ii) Cross-spectra (Equation 4.79)
$S^0(\omega)$	Cross-spectral density matrix
S_t	Strouhal number
S_c	Scruton number
$S_{ii}(f)$	Wind energy spectrum for turbulent wind components i
$S^0(\omega)$	Two-side target cross-spectral density function
$S_{ibib}(x, K)$	Buffeting force spectrum
$S_{jj}(x, K)$	Wind spectrum
$S_{Lbu}(x_1, x_2, \omega)$	Cross-spectra between the buffeting forces at two different positions x_1 and x_2
$S_{Lbu}^c(\omega)$	Auto-spectral density of the buffeting force at the center of the element
T	Time period over which the maximum value is required
T_0	Period of the sample function
U	Incoming wind velocity
$\bar{U}$	Mean wind velocity
U_α	Wind velocity with an effective angle of attack α
U_{cr}	Critical wind speed
$\bar{U}(z)$	One-hour mean wind velocity at height z
$u(t)$	Longitudinal fluctuating wind speed component
$u_j^0(t)$	One-dimensional stationary stochastic process
V	Velocity of the fluid relative to the cylinder
w_{back}	Weight factors for background response
w_{res}	Weight factors for resonant response
$w(t)$	Vertical fluctuating wind speed component
W_j	Peak inertial force in the mode j
X	Nodal displacement
$X(t)$	Displacement vector of the bridge
X_0	Amplitude of displacement vector of the bridge
$\bar{X}$	Maximum response of the structure
$\bar{X}$	Mean response of the structure
$\dot{X}$	Nodal velocity
$\ddot{X}$	Nodal acceleration
y	Vertical displacement
y_0	Vibration amplitude of the vertical displacement
y_{max}	Lock-in response of the structure
$\mathbf{Y}$	Eigenvector
$Y_1(K), Y_2(K)$	Functions of the reduced frequency K at lock-in
z	Height
z_1, z_2	Two points
α	(i) Effective angle of attack (ii) Torsional displacements of the structure (Equation 4.17)
α_0	Attack angle of mean wind speed
$\Delta\alpha$	additional incident angle induced by fluctuation
ε	Non-linear aeroelastic damping coefficient
ξ	Structural damping ratio

ξ_α	Total damping for the wind-structure system
$\xi_{0\alpha}$	Structural damping coefficient in the torsional direction
Λ	Diagonal eigenvalue matrix obtained from modal analysis
ν	Cycling rate or effective frequency for the response
μ	Dynamic viscosity of the fluid
ρ	Air density
$\rho(x)$	Correlation coefficient between the fluctuating load at position x and the load effect of interest
σ_x	Standard deviation of the structural response
σ_u	Standard deviation of wind turbulence in the along-wind direction
$\sigma_p(x)$	Standard deviation of fluctuating load at position x
$\sigma_{r,B}$	Standard deviation of the background response
$\sigma_{r,R}$	Standard deviation of the resonant response
Φ	Matrix of modal shapes
ϕ	Phase angle
ϕ_{ml}	Sequence of independent random phase angles
ϕ_i	i th natural mode shape
$\phi_1(x)$	First mode shape
$\theta_{jm}(\omega)$	Complex angle of $H_{jm}(\omega)$ for cholesky decomposition of cross-spectral density matrix $S^0(\omega)$
χ_{Du}	Aerodynamic transfer function between the horizontal fluctuating wind velocity and aerodynamic drag force
χ_{Dw}	Aerodynamic transfer function between the vertical fluctuating wind velocity and aerodynamic drag force
χ_{Lu}	Aerodynamic transfer function between the horizontal fluctuating wind velocity and aerodynamic lift force
χ_{Lw}	Aerodynamic transfer function between the vertical fluctuating wind velocity and aerodynamic lift force
χ_{Mu}	Aerodynamic transfer function between the horizontal fluctuating wind velocity and aerodynamic moment
χ_{Mw}	Aerodynamic transfer function between the vertical fluctuating wind velocity and aerodynamic moment
ω	Circular frequency of vibration
ω_f	Flutter circular frequency
ω_{ml}	Double-indexing frequency
ω_n	Structural natural frequency
ω_h	Frequency of vibration in the vertical direction
ω_s	Vortex-shedding frequency
ω_α	Frequency of vibration in the torsional direction
ω_{0h}	Structural natural frequency in the vertical direction
$\omega_{0\alpha}$	Structural natural frequency in the torsional direction
ω_{up}	Upper cutoff frequency
$\Delta\omega$	Frequency increment
Δt	Time step

References

1. Bearman, P.W. (1984) Vortex shedding from oscillating bluff bodies. *Annual Review of Fluid Mechanics*, **16**, 195–222.
2. Williamson, C.H.K. and Govardhan, R. (2004) Vortex-induced vibrations. *Annual Review of Fluid Mechanics*, **36**, 413–455.
3. Liu, H. (1991) *Wind Engineering-A Handbook for Structural Engineers*, Prentice-Hall, Inc., New Jersey.

4. Sarpkaya, T. (1979) Vortex-induced oscillations: A selective review. *Journal of Applied Mechanics*, **46**(2), 241–258.
5. Sarpkaya, T. (2004) A critical review of the intrinsic nature of vortex-induced vibrations. *Journal of Fluids and Structures*, **19**(4), 389–447.
6. Simiu, E. and Scanlan, R.H. (1986) *Wind Effects on Structures*, 2nd edn, John Wiley & Sons.
7. Ehsan, F. and Scanlan, R.H. (1990) Vortex-induced vibration of flexible bridges. *Journal of Engineering Mechanics*, **116**(6), 1392–1411.
8. Simiu, E. and Scanlan, R.H. (1996) *Wind Effects on Structures*, 3rd edn, John Wiley & Sons.
9. ASCE (1987) *Wind Loading and Wind-Induced Structural Response*, American Society of Civil Engineers.
10. Scanlan, R.H. and Tomko, J.J. (1971) Airfoil and bridges deck flutter derivatives. *Journal of the Engineering Mechanics Division*, **97**(6), 1717–1737.
11. Theodorsen, T. (1934) *General theory of aerodynamic instability and the mechanism of flutter*. NACA Report 496, National Aeronautics and Space Administration, Washington DC.
12. Agar, T.J.A. (1991) Dynamic instability of suspension bridges. *Computers & structures*, **41**(6), 1321–1328.
13. Namini, A., Albrecht, P., and Bosch, H. (1992) Finite element-based flutter analysis of cable-suspended bridges. *Journal of Structural Engineering*, **118**(6), 1509–1526.
14. Beith, J.G. (1998) A practical engineering method for the flutter analysis of long-span bridges. *Journal of Wind Engineering and Industrial Aerodynamics*, **77–78**, 357–366.
15. Ding, Q.S., Chen, A.R., and Xiang, H.F. (2002) A state space method for coupled flutter analysis of long span bridges. *Structural Engineering and Mechanics*, **14**(4), 491–504.
16. Ding, Q.S. and Xu, Y.L. (2004) 3-D Finite Element Based Flutter Analysis of Stonecutters Bridge, Report No. 03 to Hong Kong Highways Department, Department of Civil and Structural Engineering, The Hong Kong Polytechnic University, Hong Kong.
17. Chen, X.Z., Matsumoto, M., and Kareem, A. (2000) Time domain flutter and buffeting response analysis of bridges. *Journal of Engineering Mechanics*, ASCE, **126**(1), 7–16.
18. Davenport, A.G. (1962) Buffeting of a suspension bridge by storm winds. *Journal of Structural Division*, ASCE, **88**(3), 233–268.
19. Scanlan, R.H. (1978) The action of flexible bridge under wind, II: buffeting theory. *Journal of Sound and Vibration*, **60**(2), 201–211.
20. Jain, A., Jones, N.P., and Scanlan, R.H. (1996) Coupled flutter and buffeting analysis of long-span bridges. *Journal of Structural Engineering*, **122**(7), 716–725.
21. Xu, Y.L., Sun, D.K., Ko, J.M., and Lin, J.H. (2000) Fully coupled buffeting analysis of Tsing Ma suspension bridge. *Journal of Wind Engineering and Industrial Aerodynamics*, **85**(1), 97–117.
22. Bucher, G.C. and Lin, Y.K. (1988) Stochastic stability of bridges considering coupled modes. *Journal of Engineering Mechanics*, ASCE, **114**(12), 2055–2071.
23. Xiang, H.F., Liu, C.H., and Gu, M. (1995) Time-domain analysis for coupled buffeting response of long span bridge. *Proceeding of the 9th ICWE*, New Delhi, pp. 881–892.
24. Davenport, A.G. (1961) The application of statistical concepts to the wind loading of structures. *Proceedings, Institution of Civil Engineers*, **19**(4), 449–472.
25. Shinozuka, M. and Jan, C.M. (1972) Digital simulation of random processes and its applications. *Journal of Sound and Vibration*, **25**(1), 111–128.
26. Deodatis, G. (1996) Simulation of ergodic multivariate stochastic processes. *Journal of Engineering Mechanics*, ASCE, **122**(8), 778–787.
27. Ding, Q.S., Zhu, L.D., and Xiang, H.F. (2006) Simulation of stationary Gaussian stochastic wind velocity field. *Wind and Structures*, **9**(3), 231–243.
28. Lin, Y.K. and Yang, J.N. (1983) Multimode bridge response to wind excitations. *Journal of Engineering Mechanics*, ASCE, **109**(2), 586–603.
29. Xiang, H.F. (2005) *Modern Theory and Practice on Bridge Wind Resistance*, China Communications Press, Beijing (in Chinese).
30. Davenport, A.G. (1967) Gust loading factors. *Journal of the Structural Division*, ASCE, **93**, 11–34.
31. Vickery, B.J. (1966) On the assessment of wind effects on elastic structures. *Australian Civil Engineering Transactions*, **CE8**, 183–192.
32. Holmes, J.D. (2007) *Wind Loading of Structures*, 2nd edn, Taylor & Francis, Abingdon.
33. Davenport, A.G. (1964) Note on the distribution of the least value of a random function with application to gust loading. *Proceedings of the Institution of Civil Engineers*, **28**, 187–196.
34. Kasperski, M. and Niemann, H.J. (1992) The L.R.C. (load-response-correlation) - method a general method of estimating unfavorable wind load distributions for linear and non-linear structural behavior. *Journal of Wind Engineering and Industrial Aerodynamics*, **43**(1–3), 1753–1763.
35. Holmes, J.D. (1999) Equivalent static load distributions for resonant dynamic response of bridges. *Proceedings of 10th International Conference on Wind Engineering*, Copenhagen, Denmark, Balkema, Rotterdam, pp. 907–911.
36. Ding, Q.S. and Xu, Y.L. (2004) 3-D Finite Element Based Buffeting Analysis of Stonecutters Bridge, Report No. 05 to Hong Kong Highways Department, Department of Civil and Structural Engineering, The Hong Kong Polytechnic University, Hong Kong.